

CAN LANGUAGE MODELS FORECAST INDUCTIVELY?

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ABSTRACT

Judgmental forecasting involves predicting uncertain events like elections or geopolitical conflicts where abundant historical data is lacking. A key challenge is *induction*: forecasters must generalize from sparse evidence and contextual knowledge to infer how available evidence predicts future outcomes. Here, we investigate whether language models (LMs) recover and use such predictive relationships when forecasting, rather than relying on base rates or surface cues. Across a battery of experiments spanning real-world prediction markets and controlled synthetic environments, we test whether models: 1) update their beliefs coherently; 2) correctly recover predictive relationships, and 3) transfer what they learn to new settings. Our results provide evidence for each capability. Models selectively update their forecasts in response to causally relevant evidence, use contextual information to infer predictive relationships when observations are sparse, and revise these inferences as direct evidence accumulates. Training on families of predictive relationships improves forecasting across new domains, although evidence for transfer to unseen predictive structures is weaker. Together, our experiments provide behavioral evidence that LMs can forecast inductively.¹

1 INTRODUCTION

Judgmental forecasting is a demanding form of reasoning central to decisions about political, social, and economic futures. Unlike settings with abundant historical data for statistical extrapolation, judgmental forecasting requires integrating sparse, noisy evidence with contextual knowledge to form calibrated beliefs about future outcomes (Tetlock & Gardner, 2015; Mellers et al., 2014; Lawrence et al., 2006). This creates an inductive challenge: forecasters must infer from limited observations the relationships that govern how evidence impacts outcomes, and use those relationships to anticipate how events might unfold (Tenenbaum et al., 2006; Griffiths et al., 2010; Tenenbaum et al., 2011; Lake et al., 2017). Figure 1 illustrates this identification problem in a toy setting. A researcher observes a carnival coin-flipping machine and, after seeing two heads, must estimate the probability that the next flip will be heads. A judgmental forecast may incorporate contextual knowledge (e.g., that such games are often rigged) to infer the data-generating process.

Language models (LMs) now approach crowd accuracy in judgmental forecasting (Halawi et al., 2024; Schoenegger et al., 2024), but accurate predictions need not imply recovery of the underlying task structure. Models recover latent structure in some settings (Li et al., 2023), yet can predict accurately using relationships that fail to generalize in others (Vafa et al., 2025). This distinction matters when decision-makers ask how and whether a forecast should change under alternative evidence or events.

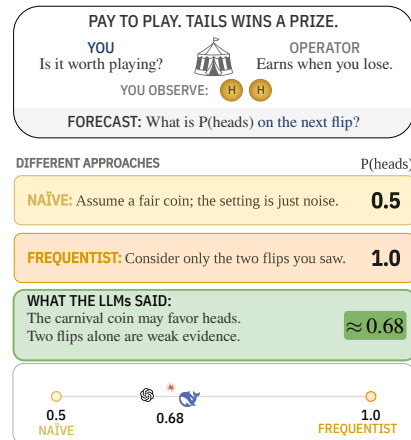


Figure 1: **Motivation.** Judgmental forecasts combine observations with contextual knowledge about the process that generated them. Frontier models, asked to forecast this question, average $P(H) \approx 0.68$ (see App. A).

¹ 🤖 Study dataset and artifacts 📄 Code repository Links anonymized for review.

Present work. We study whether LMs exhibit inductive reasoning in judgemental forecasting tasks through a series of progressively more stringent behavioral tests spanning both real-world prediction markets and controlled synthetic environments. First, we evaluate whether LMs respond selectively to new evidence, distinguishing outcome-relevant information from topically related but orthogonal information (Exp. 1). Second, we test whether LMs infer predictive relationships from described context before statistical evidence accumulates. We construct synthetic forecasting problems in which the numerical observations are identical across conditions and only the described setting varies, isolating the contribution of background knowledge to the LM’s forecast (Exp. 2). Finally, we test whether relationships learned during fine-tuning transfer to new settings. We fine-tune LMs with reinforcement learning from verifiable rewards (RLVR) on both synthetic tasks and historical prediction market data, diagnose component failures with a supervised pilot, and evaluate on held-out structures and domains (Exp. 3, Exp. 4).

Results. Across experiments, we find behavioral evidence that models forecast inductively.

- When forecasting unresolved prediction-market questions, all nine models in the magnitude analysis selectively incorporate new evidence, revising their forecast probabilities $1.9\text{--}7.5\times$ more for outcome-relevant than for merely topically related information (Exp. 1).
- In controlled synthetic worlds, we find that models use context to select between predictive relationships even when the numerical evidence is held constant, and increasingly rely on direct observations as they accumulate, as a forecaster capable of inductive reasoning should (Exp. 2). We further find that larger models can learn a new relationship from examples within the prompt. For example, when told that a city resembles either KIV or ZOR (random strings), 12 of 15 tested models can infer from examples how each city type responds to events in the simulated environment. For Qwen3-14B, manipulating the internal representation of this inferred relationship (resemblance to KIV or ZOR) changes the model’s forecasts, providing causal evidence that the inferred relationship guides its prediction.
- Finally, we find that post-training can produce forecasting behavior that generalizes across domains, although evidence for more demanding forms of transfer is substantially weaker. On synthetic data, post-training transfers robustly across a change in domain, but evidence for transfer when both the domain and the relationship between evidence and outcomes change is weaker: for Qwen3-8B, the episode-matched advantage is 0.126 (90% bootstrap CI [0.018, 0.242]) across eight seeds (Exp. 3). On 318 held-out real-world prediction markets, training on other event families reduces mean forecasting error at every tested model size, with several trained checkpoints reaching contemporaneous crowd performance (Exp. 4).

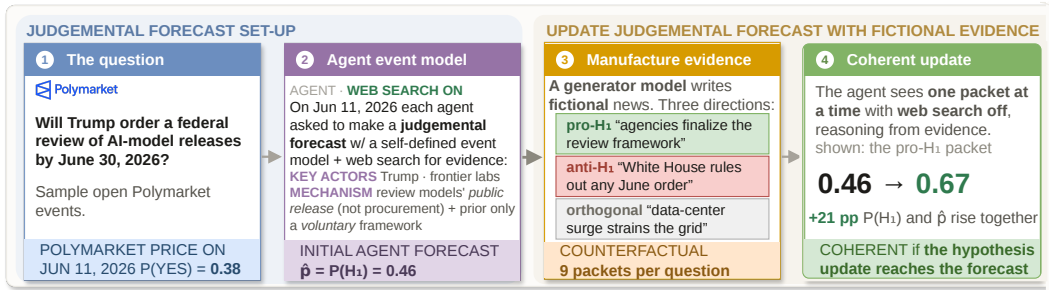


Figure 2: **When should a forecast change?** Effective forecasting requires updating one’s beliefs appropriately as new evidence arrives. This requires determining whether the evidence affects the outcome and, if so, by how much. We study these two challenges in Exp. 1 and Exp. 2, respectively. The figure illustrates our Exp. 1 setup. A model is first asked to evaluate an unresolved Polymarket question (1-2), and then we introduce synthetic new evidence that either supports the outcome, opposes it, or concerns the same topic but is largely orthogonal to it (3). Finally, with search disabled, we ask the model to revise its forecast, allowing us to measure whether it updates selectively and in the direction implied by the evidence.

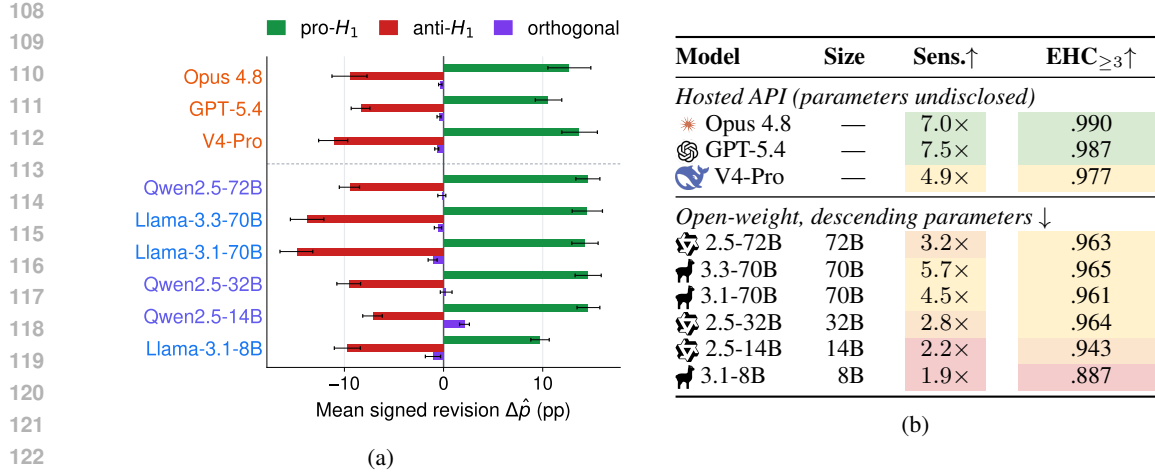


Figure 3: **Models move much more for evidence than for topical chatter, and usually move in the implied direction.** (a) Average revision after supporting, contrary, and topical updates. (b) The table summarizes how much farther models move for directional evidence conditional on moving (Sens.), and how often a move of at least three percentage points has the right sign ($\text{EHC}_{\geq 3}$). Error bars in (a) are 90% market-bootstrap intervals; robustness checks are in App. B.3.4.

2 DO FORECASTS CHANGE FOR THE RIGHT REASON?

We argue that effective updating requires a forecaster to determine: (1) which evidence is relevant in predicting an outcome and (2) the extent to which the new evidence should affect the forecast. Figure 2 illustrates these challenges considering the question of whether the Trump administration will order a federal review of new AI models. For (1), new evidence that corroborates plans for such a review should change the forecast, whereas other related AI news should not do so (unless there is a plausible causal pathway connecting it to the outcome).

We assess these capacities in two experiments. First, we pair real-world prediction-market questions with controlled synthetic evidence to test whether models update selectively and in the direction the evidence warrants. Second, we use a controlled synthetic environment where the same evidence has, by construction, different predictive effects across contexts. This lets us test whether models can infer the relationships determining how strongly the evidence should affect forecasts.

2.1 EXP. 1: CAN MODELS UPDATE IN THE RIGHT DIRECTION?

Design. We evaluate ten model deployments on 100 unresolved binary Polymarket questions with web search; nine enter the revision-magnitude analyses below. Then, we ask the models to revise their forecast after a fictional news update with search disabled (Fig. 2). Unbeknownst to the models, we construct these updates to support the outcome, oppose it, or cover the same topic without directly influencing the outcome. Because models could potentially distinguish directional from orthogonal evidence using superficial textual cues, we additionally evaluate a word-only, bag-of-words classifier that receives only the text of the news update. We also construct matched forecasting scenarios in which the same evidence should produce opposed outcomes; for example, intense rainfall could increase the probability of a good harvest in a dry region but decrease it in a region prone to flooding. (See App. B for details.)

Finding: models selectively update in the direction implied by the evidence. Across the nine models considered, and 30,094 valid forecast revisions, we find that pro-outcome evidence increases forecasts, anti-outcome evidence decreases them, and topically related but orthogonal evidence produces little change (Fig. 3a). Conditional on changing their forecast, we find that models move their probability 1.9–7.5 $\times$ farther for updates intended to influence the prediction (Fig. 3b). Human validation, uncertainty estimates, and robustness analyses are reported in App. B.

Finding: lexical cues alone do not explain model performance. We find that bag-of-words classifiers can almost perfectly distinguish directional from orthogonal evidence (.989 accuracy),

confirming that superficial cues could explain models’ selectivity. Yet, the same classifiers are much less successful at distinguishing whether directional evidence supports or opposes the outcome (.688 accuracy). Models are considerably more directionally consistent (.887–.990), suggesting lexical cues alone cannot explain their performance. More directly, when we hold the evidence fixed and vary only its relationship to the outcome, GPT-5.6 appropriately reverses the direction of its forecast update in 85% of matched scenarios, while weaker models do so substantially less often (30–37%; App. B.5).

Finding: selective updating depends on inference-time reasoning. Having ruled out simple lexical shortcuts, we next ask what makes this capability possible. In particular, does the model simply recognize a familiar pattern, or does it need inference-time reasoning to combine the stated relationship with the evidence? We isolate this by holding the model weights, prompts, and test items fixed and changing only whether reasoning is enabled. We find that Qwen3-32B’s paired-reversal success under base wording rises from 17.5% to 96.3%, while mean absolute updated-probability error falls from 26.89 to 0.08 percentage points. Since the two conditions see identical items, this gap isolates the reasoning setting rather than item difficulty or information available in the prompt (App. B.6).

Take-away. Models distinguish predictive from merely topical evidence and generally update in the implied direction. Performance drops sharply when inference-time reasoning is disabled, suggesting that this behavior depends strongly on the reasoning setting.

2.2 EXP. 2: CAN MODELS INFER HOW STRONGLY EVIDENCE SHOULD MATTER?

We study whether models can determine *how much* new evidence should update a forecast in a synthetic scenario we dub *Coin City*. We construct two reference cities (Cities A and B) where news coverage affects election polling differently: strongly in one city and weakly in the other. Specifically, an election poll in a city changes based on whether the news coverage favored one of the candidates ($x_{ij} \in [-10, -4] \cup [4, 10]$) and the city’s sensitivity to news, drawing $g_j \sim \mathcal{N}(0.90, 0.03^2)$ for the strong-response city and $g_j \sim \mathcal{N}(0.25, 0.03^2)$ for the weak-response city:

$$\Delta p_{ij} = g_j x_{ij} + \epsilon_{ij}, \quad \epsilon_{ij} \sim \mathcal{N}(0, 5^2). \quad (1)$$

We then ask the model to forecast the effect of campaign news on polling in a third city, City C, which we construct to have either the strong or weak response. Across conditions, we incrementally provide the model with additional information, asking it each time to forecast how a new piece of campaign news will affect election polling in City C. Specifically, we provide

- Up to four previous observations from City C, showing how polling numbers changed in response to news.
- Additionally, four observations from each of Cities A and B, showing how their polling numbers changed in response to news.
- Additionally, a contextual cue: strong-response cities are described as primarily receiving national news coverage and weak-response, local news coverage.
- Building on (b), we provide an *incorrect* contextual cue, linking City C to the wrong type of city.
- Finally, we replace the semantic cues with *arbitrary* labels (KIV and ZOR), requiring the model to infer their meanings.

This design allows us to test whether models use context to infer how strongly new evidence should affect their forecasts, and whether they increasingly rely on direct evidence from City C as it becomes available—all with ground truth on how forecasts should update. We evaluate six frontier systems: Claude Opus 4.8, Opus 5, GPT-5.6, DeepSeek V4-Pro, Kimi K3, and Gemini 3.6 Flash. As baselines, we also fit

Prompt: City C has a new starting poll of 53.9 and has net news of +5. What do you predict the poll will be at the end of the week?			
CITY C · being forecast prior samples			
Start	News	Change	End
57.8	+8	+9.4	67.2
CITY A national news			
Start	News	Change	End
45.4	-8	-6.5	38.9
45.4	+8	+12.2	57.6
CITY B local news			
Start	News	Change	End
35.0	-10	-5.6	29.4
35.0	+10	+9.3	44.3
HINT City C residents generally encounter campaign developments through national news coverage .			

Figure 4: To forecast polling in City C, the model may receive: (a) observations from City C, (b) reference cities A and B, and contextual cues linking City C’s response pattern to A or B (c-e)

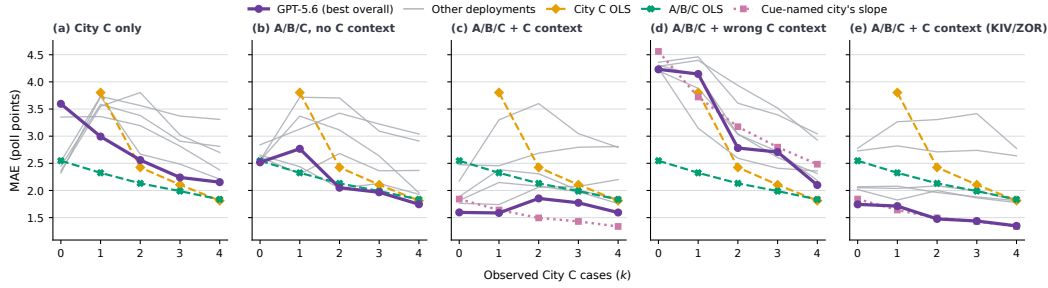


Figure 5: **Forecast error across experimental conditions.** Mean absolute error (MAE) as the number of observed City C cases increases across conditions (a)–(e). Colored lines show GPT-5.6 and OLS baselines; gray lines show the other models. Full results are in App. C.5.

OLS regressions to the observations available in each condition to estimate the relationship between campaign news and polling changes.

Finding: models use context to make better forecasts. For stronger models, this “hint” substantially improves forecasting over relying on City C alone or pooling the reference cities without knowing which one is relevant; e.g., in Fig. 5c, GPT-5.6 Sol substantially outperforms the baselines when $k = 0$ or $k = 1$. This holds true even when the cues given are arbitrary KIV/ZOR labels whose meaning must be inferred (Fig. 5e). Finally, giving the model the *wrong* contextual cue sharply worsens its forecast (Fig. 5d).

Finding: models revise their initial contextual inference as direct evidence accumulates. We find that context matters most when direct evidence from City C is sparse. As observations from City C accumulate, forecasts increasingly approach the OLS baselines. This suggests the models shift from contextual information toward direct evidence as its evidential value grows, as a forecaster should. This pattern holds across conditions in Fig. 5, but is most striking in Fig. 5d: even when initially misled by an incorrect contextual cue, models recover as direct evidence accumulates.

Mechanistic analysis. The above behavioral results show that models can infer which predictive relationship applies, *but they do not tell us whether the model represents that relationship internally or uses it to produce the forecast.* We fit linear probes to hidden states from Qwen3-14B (Yang et al., 2025) and Llama-3.1-70B (Grattafiori et al., 2024) to decode whether the model associates City C with the strong- or weak-response regime. We then test whether this representation influences the forecast using held-out matched KIV/ZOR prompt pairs. At model-specific layers, we swap the label-token hidden states (a patch) with those produced by the opposite-label prompt and regenerate the forecast. We perform this intervention both before and after City C provides direct observations.

Finding: inferred relationships are represented internally and can shape forecasts. Fig. 6 shows the effect of swapping the KIV/ZOR representation on Qwen3-14B’s forecasts when no direct City C observations are available. Before the intervention, forecasts reflect the response regime indicated by the original label; for example, prompts identifying City C as KIV produce forecasts consistent with the strong-response regime, while those identifying it as ZOR produce forecasts consistent with the weak-response regime (open circles in the figure). After swapping the internal representation associated with KIV for that associated with ZOR

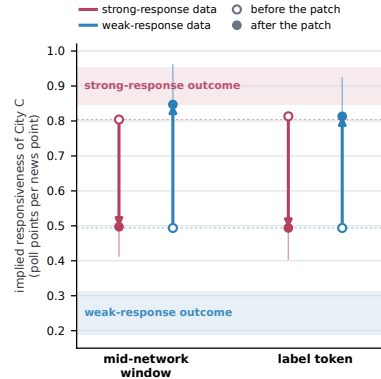


Figure 6: **Effect of representation patching on Qwen3-14B forecasts.** Predicted poll change per news point for City C before (open circles) and after (filled markers) replacing the hidden state produced by the KIV/ZOR label with that produced by the opposite-label prompt. Arrows connect predictions before and after patching; vertical lines are 90% episode-bootstrap intervals for the post-patch mean; shaded regions indicate the strong- and weak-response regimes.

(and vice versa), forecasts shift toward the opposite response regime (filled circles), moving about 51% of the way toward the swapped regime. After four direct City C observations, however, the same intervention has no reliable effect. Together, these results suggest that the inferred relationship influences the forecast when direct evidence is sparse but has less influence as City C evidence accumulates. We observe the same zero-shot causal effect with Llama-3.1-70B at a pre-specified window on fresh episodes (App. C.7.5).

Takeaway. Models can infer which predictive relationship applies, use it when direct evidence is sparse, and revise it as local evidence accumulates. In stronger models, this behavior persists under arbitrary labels. Finally, we can decode what the model infers from its internal representations, and intervening on those representations causally shifts the resulting forecast.

3 CAN MODELS LEARN TRANSFERABLE FORECASTING SKILLS?

A central idea in the forecasting literature is that forecasting is a *transferable skill*: successful forecasters rely not only on domain expertise but on flexible reasoning strategies that generalize across problems and maintain high accuracy across questions spanning diverse topics (Tetlock, 2005; Mellers et al., 2015). Specifically, inductive forecasting skills should (1) generalize across *domains*, allowing forecasters to reason effectively about unfamiliar topics, and (2) generalize across *predictive relationships*, allowing them to determine how evidence relates to outcomes rather than relying on previously learned associations.

We do so in two experiments. In Exp. 3, we use a controlled synthetic setting to independently vary domains and predictive relationships, testing whether forecasting skills acquired through post-training transfer when either changes. In Exp. 4, we move to real-world forecasting and test whether post-training on historical prediction-market questions improves forecasts on newer questions from event families and question types not encountered during training.

3.1 EXP. 3: DO LEARNED FORECASTING SKILLS TRANSFER TO NEW DOMAINS AND RELATIONSHIPS?

Design. We consider two domains, *Coin City*, the setting used for Exp. 2, and *Coin Harbor*, a new setting where shipping bulletins predict cargo flow. Coin Harbor has its own vocabulary and time unit, but uses the same distributions with different numerical values, e.g., outcomes lie on a $[40, 320]$ scale and inputs, shocks, and noise are correspondingly larger. Independently, we vary the relationship between inputs and outcomes. In Structure A, an input affects the outcome only when it occurs, similar to Exp. 2, while in Structure B an input can have effects that persist after the news signal disappears. Crossing the two domains with the two relationships yields four evaluation conditions (see Fig. 7 margins): (1) in-distribution, (2) domain shift, (3) relationship shift, and (4) joint domain-and-relationship shift.

We post-train Qwen3-8B and Qwen3-4B (Yang et al., 2025) using RLVR with Dr. GRPO (Liu et al., 2024) (details are in App. F). We train each model exclusively on Coin City episodes governed by Structure A (*episode-matched* condition). We then evaluate the trained model under all four combinations of domain and predictive relationship: Structure A in Coin City (in-distribution), Structure A in Coin Harbor (domain shift), Structure B in Coin City (relationship shift), and Structure B in Coin Harbor (joint shift). Comparing this model with the untrained *base* LM tests whether forecasting behavior acquired in one domain and predictive relationships transfer when the domain, relationship, or both change.

However, improvement over the base model may reflect learning general regularities of the training task rather than the information contained in each episode. We thus construct a second post-trained condition using the same prompts and forecast targets, but shuffling the targets across episodes. We call this the *population-prior* condition, and again evaluate it under all four combinations of domain and predictive relationship. Comparing the *episode-matched* and *population-prior* conditions isolates the benefit of learning the mapping between episode-specific evidence and forecasts, beyond learning population-level regularities of the training task.

Finding: forecasting improvements persist under a change of domain. Both the *population-prior* and *episode-matched* LMs substantially improve over the untrained base when evaluated in Coin

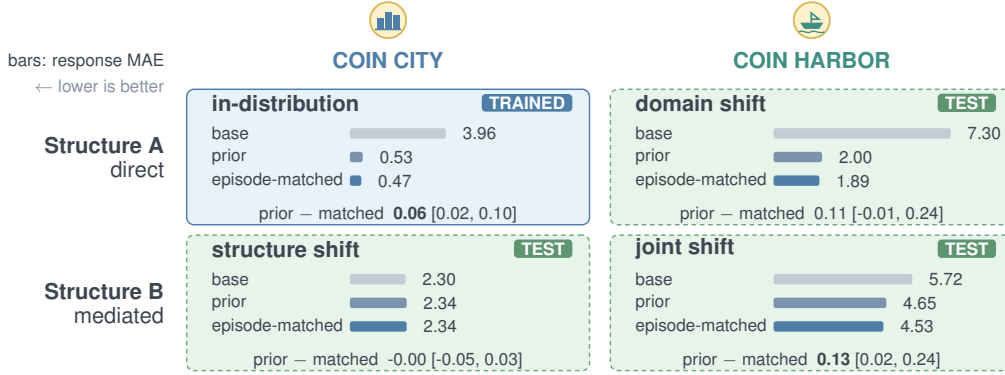


Figure 7: **Forecasting gains transfer strongly to a new domain, but not a new structure.** Mean absolute error (MAE; lower is better) for Qwen3-8B across the 2×2 combination of domain (Coin City vs. Coin Harbor) and predictive relationship (Structure A vs. Structure B). Models are trained only on Coin City with Structure A (top left) and evaluated in-distribution, under a domain shift (top right), a structure shift (bottom left), and both shifts jointly (bottom right). We compare the base model, a population-prior control trained with shuffled targets, and episode-matched training with the correct target for each episode.

Harbor, showing that training transfers useful forecasting behavior to a new surface domain (Fig. 7 top right). Under domain shift, response MAE falls from 7.30 for the untrained base to 2.00 for the *population-prior*, with all eight training seeds showing improvement over the base (Fig. 7). Thus, most but not all of the gain comes from learning the forecasting task and its population-level regularities. We then ask the stricter question of whether preserving which forecasts belong to which episode provides an additional benefit. For Qwen3-8B, the *episode-matched* LM improves over the *population-prior* LM by 0.11 response-MAE points under domain shift, with six of eight seeds favoring the episode matching setting.

Takeaway. In open-weight models, we find evidence that forecasting improvements in synthetic scenarios transfer across domains, but training under a specific predictive structure does not clearly improve performance on unseen predictive structures.

3.2 EXP. 4: CAN MODELS TRANSFER LEARNED FORECASTING SKILLS IN THE REAL WORLD?

Design. Finally, we test whether forecasting improvements from post-training extend from synthetic settings to real-world forecasting questions. We post-train Qwen3 models from 1.7B to 32B parameters and Llama-3.1-8B on 1,736 resolved historical Polymarket questions and evaluate them on 1,024 held-out questions and, for the scale comparison below, a further 318-market holdout disjoint from both. For each market, we construct a forecasting task 30 days before resolution, providing the model with the question, resolution criteria, and pre-cutoff market-price history. Without access to external information, the model produces a probability forecast for the eventual binary outcome.

To test generalization, we make the train and test sets disjoint along two dimensions: test questions concern entirely different underlying events and follow different question templates, requiring models to transfer across both real-world contexts and question structures. We evaluate forecasting accuracy using the Brier score, where lower values indicate more accurate forecasts (App. E). As external benchmarks, we compare model forecasts with contemporaneous Polymarket prices on the same questions (which correspond to forecasts) and with forecasts from four frontier models without post-training.

Finding: post-training improves forecasting on unseen real-world questions, with diminishing gains for stronger base models. We find that post-training reduces forecasting error across the tested Qwen3 models, but this improvement is substantial only for the smaller models (Fig. 8a). For instance, Qwen3-1.7B improves from a Brier score of .225 to .125, while Qwen3-8B improves from .137 to .129. As the untrained models become stronger, however, the gap narrows such that at 32B,

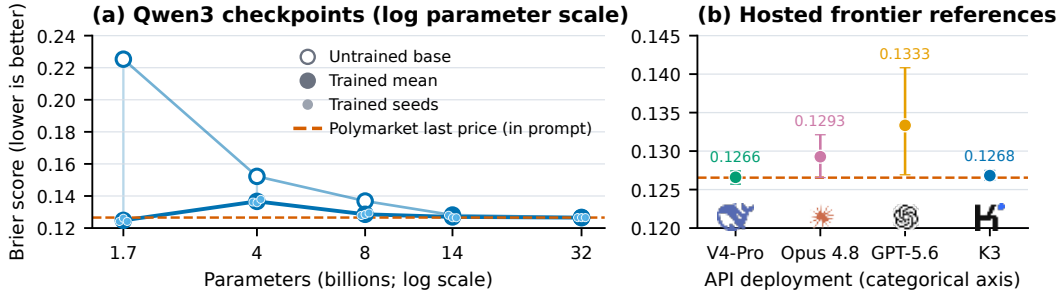


Figure 8: **Post-training improves forecasting accuracy, with gains diminishing as base-model performance improves.** (a) Brier scores for Qwen3 models from 1.7B to 32B parameters before and after post-training on historical prediction markets (lower Brier scores indicate more accurate forecasts). Post-training produces the largest improvement at 1.7B, with progressively smaller differences between trained and untrained models at larger scales. Several post-trained models reach approximately the Brier score of contemporaneous Polymarket prices (dashed line). Points show individual training seeds. (b) Brier scores of four hosted frontier models evaluated on the same held-out questions, with 90% family-clustered bootstrap intervals. Their performance falls in a similar range to the strongest post-trained Qwen3 models and the Polymarket benchmark. The Llama replication is reported in App. E.4.

performance is roughly the same (.1265 versus .1266). The Llama replication shows the same broad pattern; see App. E.4.

Finding: post-trained models reach prediction-market and frontier-model forecasting accuracy. Several trained LMs also reach approximately the same Brier score as contemporaneous Polymarket prices on the shared holdout. Their performance is also comparable to the four untrained frontier models we evaluate: the frontier models obtain Brier scores between .1266 and .1333, similar to several of our post-trained Qwen3 models (Fig. 8b). Thus, the forecasting gains learned during post-training are not confined to our synthetic environments; they extend to event families and questions absent from training. At the same time, the diminishing base-relative gain suggests that stronger pretrained models already possess more of the forecasting behavior that post-training teaches weaker models.

Takeaway. Using data from real-world prediction markets, we find evidence that post-training improves forecasting beyond synthetic settings, where we can directly observe predictive relationships. We find that gains are largest for weaker base models and diminish as the untrained forecaster becomes stronger. A price-history ablation shows that this accuracy depends strongly on the supplied market history (App. E.5). Because this experiment compares trained models only with their untrained bases, however, it establishes improved out-of-distribution forecasting accuracy rather than isolating episode-specific relationship learning.

4 RELATED WORK

Judgmental forecasting. Work on human judgmental forecasting has identified base rates, decomposition, evidence aggregation, and continuous belief revision as consistent characteristics of strong forecasters (Tetlock, 2005; Mellers et al., 2014; Tetlock & Gardner, 2015). More recently, work on LMs has asked how well these systems can forecast real-world events, developing leakage-resistant benchmarks and improving accuracy through retrieval, ensembling, prompting, and post-training (Zou et al., 2022; Karger et al., 2025; Halawi et al., 2024; Schoenegger et al., 2024). This literature has established increasingly strong forecasting performance, in some settings approaching or beating the accuracy of human crowds. Related work has also begun to examine whether LMs employ established forecasting strategies (Pratt et al., 2024). We complement this literature by examining a different dimension of forecasting capability: whether LMs can forecast inductively, identifying predictive relationships from evidence and using them to revise their forecasts.

Induction and relational reasoning. Transformers can infer functions from examples, and work on analogy and world models have investigated whether learned representations capture relations

or dynamics (Garg et al., 2022; Chan et al., 2022; Webb et al., 2023; Yasunaga et al., 2024; Ha & Schmidhuber, 2018; LeCun, 2022). Evidence on this question is mixed. In some settings, models appear to recover structured representations of the processes generating their observations, such as the state of an Othello board (Li et al., 2023). In others, successful prediction can coexist with representations that fail to capture underlying spatial or temporal structure (Vafa et al., 2025). More broadly, this reflects the problem of *shortcut learning*, where models can perform well by exploiting predictive associations without recovering the underlying structure that generated them (Geirhos et al., 2020). This distinction is particularly important for forecasting, as producing an accurate prediction does not necessarily imply that a model has recovered the predictive relationships that make the available evidence informative. We study this distinction behaviorally, testing whether LMs can infer such relationships from evidence, use them to revise their forecasts, and transfer them when the setting changes.

Simulating human behavior. A related literature uses LMs to simulate human behavior, asking whether synthetic agents can reproduce the responses of individuals or populations (Park et al., 2023; Aher et al., 2023; Argyle et al., 2023). While early work emphasized agreement between simulated and observed outcomes, more recent work has questioned whether matching aggregate behavior is sufficient: models can reproduce population-level outcomes while failing to preserve relationships between variables within the population (Bisbee et al., 2024; Xie et al., 2026). Forecasting raises an analogous problem, as models may produce accurate probabilities without capturing the predictive relationships that connect evidence to outcomes. Our work studies this form of conditional validity in forecasting, asking not only whether models predict accurately, but whether their forecasts change appropriately as the evidence and setting change.

5 DISCUSSION & LIMITATIONS

Our experiments provide behavioral evidence that LMs can forecast inductively, inferring, using, and revising predictive relationships as evidence changes. These behaviors do not establish human-like understanding or general world modeling capabilities, but nonetheless show that models behave in ways we expect of competent human forecasters. This distinction matters as LM forecasts may increasingly inform real-world decisions (Schoenegger et al., 2025), where accuracy in familiar settings provides limited reassurance if forecasts become brittle when the evidence, context, or underlying predictive relationships change.

This may reflect the limited training environment, and these capabilities may emerge with enough variation in the predictive structures encountered during training. The contrast between Exp. 2 and Exp. 3 points to a possible asymmetry. Models flexibly infer and revise predictive relationships from evidence available in context (Fig. 5), while relationships acquired through post-training transfer less reliably when the underlying structure changes (Fig. 7). Although these experiments are not directly comparable, understanding when relationships inferred in context can become portable capabilities through post-training is an important direction for future work.

Limitations. Our synthetic environments deliberately simplify real forecasting to isolate predictive relationships, and our mechanistic analyses are limited to open-weight models. We also do not study agentic forecasters that can actively search for information, test competing hypotheses, or seek evidence that distinguishes among possible predictive relationships. Such capabilities may change the induction problem by allowing models to determine which relationship applies rather than relying only on the evidence provided in context. Exp. 4 extends our evaluation to real-world forecasting, but improved held-out accuracy does not establish that these gains arise from the episode-specific relationship learning isolated in our controlled experiments.

AI USE STATEMENT

Language models produced the forecasts and experimental responses analyzed in this work and assisted with software and manuscript preparation. They also generated the context-reversal development vignettes. The authors verified numerical claims against saved experimental records and executable analyses, checked citations against original sources, reviewed the code, and take responsibility for the submission and its artifacts.

ETHICS STATEMENT

The reported analyses use public market metadata and price histories, model outputs, synthetic worlds, and an unpaid adult review of synthetic materials. The review collected no identifying or sensitive information. Under our institutional guidelines, it does not constitute human-subjects research; consequently, no IRB review was sought. Counterfactual evidence packets were evaluated offline and were not posted to markets or used for trading. We report aggregate model behavior rather than recommend deployment.

REPRODUCIBILITY STATEMENT

The appendices specify the prompts, data-generating processes, sample construction, censoring, estimators, deployment settings, missingness, hyperparameters, and baselines needed to reproduce the reported analyses. Builders, manifests, checksums, and saved generations support each result. Experiment 3 reports the eight-seed Qwen transfer comparison, a four-seed supervised learnability diagnostic, and five-seed gain-intervention and shuffled-target analyses; later outcome-informed extensions are labeled descriptive in the appendix. Experiment 4 evaluates Qwen and Llama checkpoints on family-disjoint historical markets, with a separate scale holdout and price-history ablation. Evaluation examples are excluded from training; selection rules and exploratory analyses are documented in the appendix.

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APPENDICES

The appendices document the experimental designs, analyses, and supplementary results. Appendix A describes the coin-flip illustration in Figure 1. Appendices B–E present the methods and results for Experiments 1–4: forecast updating, contextual relationship selection, controlled transfer, and historical-market forecasting. Appendix F gives the shared training configuration for Experiments 3–4.

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A DESCRIPTIVE COIN-FLIP FORECAST ELICITATION

Figure 1 illustrates a simple forecasting problem: predicting the next flip of an unfamiliar coin after observing two heads. We elicited eight independent forecasts from each of GPT-5.4, Claude Opus 4.8, and DeepSeek V4-Pro at temperature 0.7 with a 600-token output limit. All 24 responses contained a parseable probability. Each model marker in Figure 1 shows its eight-call mean, and the displayed $\approx .68$ value rounds the pooled mean of .679. This descriptive probe is separate from Experiments 1–4.

The exact user prompt was:

You are at a carnival game. A coin you have never seen before is flipped twice in front of you, and it comes up heads both times. These two flips are everything you know about this coin.

What is the probability that the next flip comes up heads?

Think about it intuitively, the way you actually would in the moment -- do NOT use any formula, rule, or calculation, and do not work out any numbers in your reasoning. Just reason in plain words about what you'd expect and why. Then END your reply with your gut answer as a JSON object on its own line:

```
{"rationale": "one short sentence, no math",
  "p_heads": <number between 0 and 1>}
```

Table 1: Forecasts underlying the three model markers in Figure 1. Each row summarizes eight independent temperature-0.7 calls.

Model	Mean	Median	Range
GPT-5.4	.644	.670	[.60, .67]
Claude Opus 4.8	.681	.675	[.65, .75]
DeepSeek V4-Pro	.712	.750	[.60, .75]
Pooled ($n = 24$)	.679	.670	[.60, .75]

The probe contains no matched control condition. The prompt describes an unfamiliar carnival coin and requests intuitive reasoning without calculation, so the observed forecasts cannot distinguish the effect of the carnival context from ordinary updating about an unknown coin bias. The 0.5 and 1.0 endpoints in Figure 1 are illustrative reference policies rather than measured control responses. For comparison, Laplace’s rule of succession gives a next-flip probability of $3/4$ after two heads under a uniform prior over the coin’s unknown bias. The pooled model mean of .679 lies between the fair-coin reference of .5 and this Bayesian benchmark. Accordingly, the probe motivates the distinction between contextual knowledge and observed evidence without identifying their separate contributions.

B EXPERIMENT 1: PROSPECTIVE FORECAST UPDATING

Experiment 1 tests whether forecast revisions follow the direction implied by new evidence and whether outcome-relevant evidence produces larger revisions than topically related but outcome-orthogonal information. We evaluated ten model deployments. Nine enter the primary revision-magnitude comparisons, while Qwen2.5-7B is retained only for directional-consistency analyses because some of its recorded updates mix probability scales (Section B.3.1). Unless stated otherwise, sample sizes refer to parseable records in the June 17, 2026 analysis dataset.

B.1 STUDY DESIGN, SAMPLE, AND AGENT ROSTER

B.1.1 DESIGN OVERVIEW

Each agent first forecast unresolved binary Polymarket questions. For each of 100 selected markets, we constructed nine synthetic evidence packets: three supporting YES (pro- H_1), three opposing YES (anti- H_1), and three designed to be topically related but outcome-orthogonal. Each update paired one packet with the agent’s own initial forecast, and packets were presented independently so revisions did not accumulate across interventions. The unit of analysis is an *agent* $\times$ *market* $\times$ *initial forecast* $\times$ *evidence packet* update. The design requested five initial forecasts per market from hosted systems and three from locally served systems. Observed sample sizes vary because some initial forecasts or updates were unavailable or unparseable. Section B.3 defines eligibility and aggregation for each estimand.

B.1.2 SAMPLING FRAME AND MARKET SELECTION

The sampling frame is a Polymarket API snapshot taken on June 9, 2026. Eligible markets were binary, active, and unresolved; scheduled to resolve 14–180 days after sampling; priced strictly between .10 and .90 on the YES contract; and supported by at least \$1,000 in volume. A fixed keyword filter retained politics, government, elections, economics, technology, and geopolitics while excluding sports, cryptocurrency, entertainment, and other out-of-scope topics. When multiple contracts belonged to the same event, we retained the highest-volume contract.

The filters reduced 60,989 binary records to 37,386 active markets, 8,639 markets after topic filtering, 3,570 after the resolution-window restriction, 1,403 after the price restriction, and 839 after the volume requirement. Event-level deduplication left 436 candidates. We selected 100 markets in descending order of trading volume, admitting a candidate only when its Jaccard similarity to every previously selected market was below .25 and fewer than 16 markets had already been selected from its topic category. Similarity was computed from unigram and bigram sets derived from the question and event title after lowercasing and stopword removal. The final sample contains 16 global-election, 16 other, 15 Iran-nuclear, 12 U.S.-federal-policy, 10 AI/technology, 9 U.S.-state-election, 7 Russia-Ukraine, 5 Brazil, 5 U.S.-midterm, 3 macroeconomic, and 2 China-Taiwan markets.

All deployments were evaluated on this same 100-market sample, subject to response availability. Llama-3.1-70B has 99 analyzable markets because one market lacks a parseable initial forecast. Claude Opus 4.8 has usable updates in all three evidence conditions for 73 markets. Section B.4.3 examines robustness to these coverage differences.

B.1.3 AGENT ROSTER AND EXECUTION MODES

The study includes three hosted API systems and seven locally served open-weight systems. Hosted systems used Azure AI Foundry’s native search-supported interface (Microsoft, 2026), while local systems ran through Ollama (Ollama, 2026) and received search results from Tavily. All agents followed the same substantive forecasting procedure: construct an event model, gather current evidence, and return a structured probability forecast. Search interfaces and inference runtimes nevertheless differed across deployments.

Table 2: Experiment 1 model deployments and requested initial forecasts per market.

Model	Runtime	Search provider	Parameters	Requested k
GPT-5.4	Azure Foundry	Foundry native	—	5
Claude Opus 4.8	Azure Foundry	Foundry native	—	5
DeepSeek V4-Pro	Azure Foundry	Foundry native	—	5
Qwen2.5-7B	Ollama	Tavily	7B	3
Qwen2.5-14B	Ollama	Tavily	14B	3
Qwen2.5-32B	Ollama	Tavily	32B	3
Qwen2.5-72B	Ollama	Tavily	72B	3
Llama-3.1-8B	Ollama	Tavily	8B	3
Llama-3.1-70B	Ollama	Tavily	70B	3
Llama-3.3-70B	Ollama	Tavily	70B	3

Local-model reasoning calls used temperature .1, and the final JSON-formatting call used temperature zero. Initial probabilities expressed on a 0–100 scale were converted to 0–1 before the update stage. Section B.3.1 describes the mixed-scale Qwen2.5-7B update records and their treatment.

B.2 FORECAST, INTERVENTION, AND VALIDATION PROTOCOL

B.2.1 INITIAL FORECAST ELICITATION

Initial forecasts were elicited in three turns using the same substantive instructions across deployments, with provider-specific adaptations for search and tool use. The prompt emphasizes explicit resolution criteria and the status quo, following prior forecasting-prompt work (Halawi et al., 2024; Wilson, 2025).

1. **Event model.** The agent received the question, up to 1,500 characters of resolution criteria, days to resolution, and category, and then defined YES and NO outcomes, identified relevant actors and mechanisms, formed H_1 and H_2 , and described the status-quo trajectory.
2. **Current-evidence research.** The agent gathered evidence for and against each hypothesis, assessed whether the events required for YES were plausible within the remaining time, and revised its assessment.
3. **Structured forecast.** The agent returned a structured forecast containing the event model, hypothesis probabilities, final YES probability, confidence, rationale, and evidence sources.

The research turn included the following calibration instruction:

Remember: good forecasters put extra weight on the status quo. Only depart substantially from the status quo you identified in Step 1 if you have strong, concrete directional evidence.

The output contract records both the final YES probability, $\hat{p}$, and the posterior probability assigned to H_1 , and explicitly requires these quantities to agree. The internal coherence score in Section B.3.2 measures adherence to this constraint after updating. The fields relevant to the updating analysis are:

```
{
  "hypotheses": [
    {"id": "H1", "posterior_probability": float},
    {"id": "H2", "posterior_probability": float}
  ],
  "yes_prob": float
}
```

B.2.2 COUNTERFACTUAL PACKET CONSTRUCTION AND VALIDATION

GPT-5.4 generated nine synthetic evidence packets for each market: three pro- H_1 , three anti- H_1 , and three orthogonal. Generation used the market question, abridged resolution criteria, and one reference initial event model and forecast. Every deployment received the same frozen packet set. Each packet contains a short synthetic news report together with its intended direction, evidence type,

and targeted causal mechanism. Within each direction, the generator was instructed to vary the actor, mechanism, and framing across successive packets. Search was disabled during packet generation, so the interventions are hypothetical premises rather than reports drawn from contemporaneous news.

Table 3: Evidence-packet taxonomy. Each market receives three pro- H_1 , three anti- H_1 , and three orthogonal packets, for nine interventions in total.

Slot type	Direction	Role in the intervention
DATA	pro, anti	A measurement, price signal, statistic, or poll bearing on the likelihood of H_1 .
DECISION	pro, anti	An actor decision, official statement, policy announcement, commitment, or refusal bearing on H_1 .
STRUCTURAL	pro, anti	A technical, logistical, legal, or contextual change affecting the likelihood of H_1 .
PROCEDURAL	orthogonal	A same-domain administrative or procedural development without a direct causal path to H_1 versus H_2 .
PERSONNEL	orthogonal	A staffing or leadership change involving a relevant actor but an issue unrelated to H_1 versus H_2 .
BACKDROP	orthogonal	Economic, logistical, or social context that is topically relevant but should not change the outcome probability.

All 900 intended packets were successfully generated, with 300 in each direction class. A separate human materials review evaluated 18 packets from distinct markets, balanced across the three conditions. Among eight eligible reviewers, the majority judgment matched the intended direction for 17 of 18 packets, while individual judgments matched on 108 of 144 responses. Section B.4.4 reports the full review design and results. The review supports interpretability of the intended direction on the sampled materials but does not establish that all 900 packets are unambiguous or matched in evidential strength.

B.2.3 FORECAST REVISION AND PAIRING

Each update paired one evidence packet with the agent’s own initial structured forecast. Search was disabled, and the agent returned revised H_1 , H_2 , and YES probabilities using the same output contract as the initial forecast. Each update contained exactly one initial forecast and one packet, so interventions were evaluated independently rather than accumulated across packets. For an agent–market pair with k parseable initial forecasts, the complete crossed design contains $9k$ updates. Repeated updates share markets, initial forecasts, and evidence packets, so uncertainty is clustered by market as described in Section B.3.3. Observed sample sizes differ across deployments because some forecasts or updates were unavailable or unparseable.

GPT-5.4 received one additional instruction that no other deployment received. It was asked to judge whether the packet was already reflected in its initial forecast or represented genuinely new information, with the former associated with a smaller revision and the latter with a larger one. Because this instruction directly concerns revision magnitude, GPT-5.4’s exact-zero revision rate is not directly comparable with those of the other deployments. All deployments otherwise use the same scoring definitions below.

B.3 ESTIMANDS AND AGGREGATION

B.3.1 ANALYSIS SAMPLE AND PROBABILITY SCALES

The analysis sample is defined by the frozen June 17, 2026 dataset. Evidence packets and updates are deduplicated by their identifiers, with a parseable update retained when both a valid response and an error record share an identifier. Eligibility is defined separately for each estimand, so records with different available fields can contribute to different metrics.

Qwen2.5-7B contains 207 of 8,636 deduplicated updates with a recorded YES probability above 1.5, indicating a mixture of 0–1 and 0–100 probability scales. Its absolute revision magnitudes and sensitivity ratio are therefore excluded from magnitude-based comparisons. Directional-consistency analyses instead normalize each probability field independently before scoring, allowing all ten deployments to contribute.

B.3.2 DIRECTIONAL CONSISTENCY AND EVIDENCE SELECTIVITY

Let $\hat{p}_i^{\text{initial}}$ and $\hat{p}_i^{\text{post}}$ denote the initial and revised YES probabilities for update i , and let $p(H_1)_i^{\text{post}}$ denote the revised probability recorded for H_1 . For directional packets, let $d_i = +1$ for pro- H_1 evidence and $d_i = -1$ for anti- H_1 evidence. Orthogonal packets have no intended direction and are excluded from EHC and HFC. The primary directional analyses condition on an absolute revision of at least three percentage points; Section B.3.4 evaluates alternative thresholds.

Evidence–hypothesis consistency. Define

$$\Delta p(H_1)_i = p(H_1)_i^{\text{post}} - \hat{p}_i^{\text{initial}}.$$

For directional records with $|\Delta p(H_1)_i| \geq 0.03$,

$$\text{EHC}_i = \mathbf{1}[\Delta p(H_1)_i d_i > 0]. \quad (2)$$

EHC measures whether the revised H_1 probability moves in the direction implied by the packet.

Hypothesis–forecast consistency. Define

$$\Delta \hat{p}_i = \hat{p}_i^{\text{post}} - \hat{p}_i^{\text{initial}}.$$

For directional records with $|\Delta \hat{p}_i| \geq 0.03$,

$$\text{HFC}_i = \mathbf{1}[\Delta \hat{p}_i d_i > 0]. \quad (3)$$

HFC measures whether the revised YES probability moves in the direction implied by the packet. Because the prompt explicitly requires the YES probability to equal the revised H_1 probability, HFC and EHC are not independent measures.

Internal coherence. For records with both revised fields,

$$\text{ICS}_i = \mathbf{1}[|\hat{p}_i^{\text{post}} - p(H_1)_i^{\text{post}}| < 0.02]. \quad (4)$$

ICS measures adherence to the prompted equality between the two forecast fields.

Unconditional evidence selectivity. Mean absolute revision is summarized by

$$\text{Sensitivity} = \frac{|\Delta \hat{p}|_{\text{pro+anti}}}{|\Delta \hat{p}|_{\text{orthogonal}}}. \quad (5)$$

The numerator pools pro- H_1 and anti- H_1 updates, while the denominator pools orthogonal updates. Both include zero revisions. A value of one indicates equal mean absolute movement in the directional and orthogonal conditions, while larger values indicate greater movement in response to directional evidence. Because this ratio combines revision frequency with revision magnitude conditional on moving, Section B.4.3 reports those components separately.

B.3.3 AGGREGATION AND UNCERTAINTY

For EHC, HFC, and ICS, eligible records are first pooled within each model–market pair to obtain a market-level rate. The reported model score is the unweighted mean of these market-level rates, so markets with more completed repetitions receive no additional weight. Standard errors are computed from variation across contributing markets.

Revision magnitudes and sensitivity use record-pooled point estimates. Their 90% intervals use 20,000 percentile bootstrap resamples of markets, retaining all available packets and repeated updates from a selected market within the same resample. This preserves the reported point estimand while accounting for dependence among updates from the same market.

B.3.4 OUTPUT COMPLIANCE AND THRESHOLD ROBUSTNESS

The output contract explicitly requires the revised YES probability to equal the revised H_1 probability. After scale normalization, 25,982 of 26,008 directional records with both fields present (99.90%) satisfy this equality to numerical precision, and 25,985 (99.91%) differ by less than two percentage points. ICS is therefore primarily a schema-compliance measure, and agreement between EHC and HFC should not be interpreted as independent evidence for a latent hypothesis-to-forecast mechanism.

We recompute directional correctness at movement thresholds of 0, 1, 3, and 5 percentage points. At threshold zero, every valid directional update enters the denominator and an unchanged forecast is scored as incorrect. At positive thresholds, the analysis conditions on an absolute revision of at least the stated magnitude. Rates are computed within markets and then averaged equally across markets.

The threshold sweep leaves the substantive conclusion unchanged. Across hosted deployments, unconditional directional correctness is 97.3–98.4%, compared with 97.7–99.0% at the primary three-percentage-point threshold. The three-percentage-point rule excludes 6.9% of GPT-5.4, 12.6–12.7% of DeepSeek V4-Pro, and 18.1% of Claude Opus 4.8 directional updates. For local deployments, it excludes only 0.6–2.9% of directional updates, while unconditional correctness ranges from 88.1% to 96.4%. The high conditional rates therefore do not arise from excluding most updates, although conditioning on movement modestly increases some scores. These metrics evaluate revision direction rather than calibration or optimal revision magnitude.

B.4 STUDY LIMITATIONS AND ROBUSTNESS

B.4.1 SAMPLING AND OUTCOME SCOPE

The 100-market sample comprises liquid, medium-horizon Polymarket questions selected by topic, price, resolution horizon, trading volume, event deduplication, and topical-diversity constraints. Experiment 1 therefore characterizes updating on this selected population rather than forecasting tasks generally. Experiment 4 separately evaluates resolved historical markets with aligned information cutoffs for model forecasts and market prices.

B.4.2 INFORMATION AVAILABLE IN THE UPDATE PROMPT

Agents received their own initial forecast, the synthetic evidence text, and the update instructions. The packet’s intended direction, target hypothesis, expected shift, evidence type, mechanism annotation, and generator rationale were withheld. Among 12,728 retained hosted-system prompts, every evidence block matched the corresponding packet text and contained none of this withheld metadata. None of the 900 packets explicitly names H_1 , H_2 , the prediction market, or its resolution. Twenty-two packets contain likelihood-related language, but those expressions describe events within the synthetic report rather than instructing the agent how to revise its forecast. The intended label was therefore not directly exposed, although indirect cues from vocabulary and evidence type remain possible.

B.4.3 ROBUSTNESS OF EVIDENCE SELECTIVITY

Coverage and material construction. Most deployments have usable directional and orthogonal updates for at least 99 of the 100 markets. Claude Opus 4.8 has all three evidence conditions for 73 markets, and restricting its sensitivity estimate to this common set gives $23.7\times$, compared with $24.2\times$ on its full available sample. Provider failures were more frequent for hosted systems, so these coverage checks do not rule out content-dependent missingness. Text-based robustness analyses additionally exclude 346 updates that cannot be linked to the frozen 900-packet set, while numerical analyses that require only forecast revisions retain them.

All packets were generated from a reference event model supplied by DeepSeek V4-Pro. Its evaluation is consequently less independent of packet construction than those of the other deployments. The human materials review in Section B.4.4 provides a separate assessment of packet direction on a subset of the materials.

Revision frequency and conditional magnitude. The unconditional sensitivity ratio combines two behaviors: whether a forecast changes at all and how far it moves conditional on changing. Let A

denote the fraction of updates with $\Delta\hat{p} = 0$ and let M denote mean absolute revision among nonzero updates. Then

$$\text{Sensitivity} = \underbrace{\frac{M_{\text{dir}}}{M_{\text{orth}}}}_{\text{magnitude conditional on revision}} \times \underbrace{\frac{1 - A_{\text{dir}}}{1 - A_{\text{orth}}}}_{\text{revision frequency}}. \quad (6)$$

Directional zero-revision rates are 0.0–1.4% across deployments, whereas orthogonal zero-revision rates range from 1.1% to 71.4%. After conditioning on a nonzero revision, directional evidence still produces $1.9\text{--}7.5\times$ larger movements than orthogonal evidence, compared with unconditional ratios of $1.9\text{--}24.2\times$. Thus the selectivity result reflects both a greater probability of revising and a larger revision conditional on doing so.

Table 4: Decomposition of evidence selectivity into exact-zero revision frequency and revision magnitude conditional on moving. Brackets give 90% market-bootstrap intervals. Qwen2.5-7B is excluded because its recorded revision magnitudes mix probability scales.

Model	Orthogonal	Mean $ \Delta\hat{p} $ given a revision		Sensitivity	
	zero revision	orthogonal	directional	given a revision	unconditional
Claude Opus 4.8	71.4% [65.0, 77.4]	1.6 pp	11.2 pp	$7.0\times$ [5.7, 8.7]	$24.2\times$ [18.2, 34.3]
GPT-5.4	8.4% [5.6, 11.3]	1.3 pp	9.6 pp	$7.5\times$ [6.8, 8.4]	$8.2\times$ [7.4, 9.1]
DeepSeek V4-Pro	44.4% [39.8, 49.0]	2.6 pp	12.7 pp	$4.9\times$ [4.5, 5.4]	$8.9\times$ [8.1, 9.7]
Qwen2.5-72B	24.2% [20.3, 28.1]	3.9 pp	12.6 pp	$3.2\times$ [3.0, 3.4]	$4.2\times$ [4.0, 4.5]
Llama-3.3-70B	10.7% [7.6, 14.1]	2.6 pp	14.6 pp	$5.7\times$ [5.3, 6.1]	$6.4\times$ [5.9, 6.9]
Llama-3.1-70B	2.5% [1.0, 4.6]	3.4 pp	15.2 pp	$4.5\times$ [4.2, 4.7]	$4.6\times$ [4.3, 4.9]
Qwen2.5-32B	11.1% [8.4, 14.0]	4.5 pp	12.6 pp	$2.8\times$ [2.7, 3.0]	$3.2\times$ [3.0, 3.4]
Qwen2.5-14B	26.1% [21.4, 31.0]	5.5 pp	11.9 pp	$2.2\times$ [2.0, 2.3]	$2.9\times$ [2.6, 3.2]
Llama-3.1-8B	1.1% [0.6, 1.7]	6.0 pp	11.6 pp	$1.9\times$ [1.8, 2.0]	$1.9\times$ [1.9, 2.0]

Exact-zero revision is itself an imperfect measure of relevance because it depends on output precision and the update prompt asks agents to leave unaffected fields unchanged. We therefore interpret zero-revision frequency descriptively rather than as evidence about an internal decision to ignore information.

Surface-form differences. Directional and orthogonal packets differ by construction in evidence type as well as outcome relevance. Directional packets are somewhat longer and contain more digits, but restricting both conditions to their shared 66–114-word range retains 889 of 900 packets and changes every sensitivity estimate by less than 0.1. Length alone therefore does not explain the selectivity effect.

Vocabulary does distinguish the packet classes. With five-fold cross-validation grouped by market, multinomial naive Bayes classifies directional versus orthogonal packets with .989 accuracy against a .667 majority baseline. Evidence selectivity alone therefore cannot establish that models inferred outcome relevance rather than recognizing surface features associated with the packet type.

Recovering the *direction* of the evidence is substantially harder for the same lexical baselines. On pro- H_1 versus anti- H_1 packets, naive Bayes and logistic regression achieve .688 and .675 accuracy against a .500 baseline, and adding the market question yields .665 and .680. These values remain well below the hosted deployments’ unconditional directional correctness of .973–.984. The comparison rules out these particular bag-of-words explanations for most of the sign accuracy, but it does not rule out stronger surface-based heuristics.

Cross-deployment comparisons. GPT-5.4 alone received the additional novelty instruction described in Section B.2.3, and hosted and local systems used different retrieval and inference stacks. Comparisons between individual deployments, model families, or hosted and open-weight systems are therefore descriptive rather than controlled effects of model identity or scale.

B.4.4 HUMAN MATERIALS REVIEW

Nine adult reviewers assessed 18 synthetic reports from distinct markets, balanced across pro- H_1 , anti- H_1 , and orthogonal evidence. Each item presented the market question, resolution criteria, the corresponding pre-update forecast summary, and the synthetic report while withholding the intended direction, generator metadata, and model responses. Reviewers judged the report’s effect on the outcome probability and whether it was usable as a stipulated hypothetical premise.

One reviewer was excluded under a response-pattern criterion after selecting the same direction on 17 of 18 balanced items, leaving eight eligible reviewers. Across these reviewers, 108 of 144 judgments (75.0%) matched the intended direction. The panel-majority judgment matched on 17 of 18 items (94.4%; Cohen’s $\kappa = .919$), and reviewers judged the report usable as a premise on 124 of 144 judgments (86.1%). Including the excluded reviewer gives 115 of 162 judgments (71.0%) matching the intended direction.

The review supports the interpretability of the intended direction on the sampled materials while showing that individual judgments were not unanimous. It is a materials-validity check rather than a matched human forecasting baseline, since reviewers classified 18 reports whereas agents revised numerical forecasts across the full intervention set.

Scope of the intervention. Orthogonal packets provide topically related information intended not to affect the outcome, but they are not equivalent to receiving no new information. Experiment 1 therefore cannot separate generic effects of re-eliciting a forecast from responses to topically related orthogonal text. The update task also instructs agents to treat each synthetic report as a supplied premise and disables external search, so it measures conditional evidence use rather than source verification or whether the report should be trusted.

B.5 CONTEXT REVERSAL WITH FIXED NEWS

The primary intervention changes both outcome relevance and evidence type, leaving open whether models can infer relevance when the news itself is held fixed. An exploratory context-reversal task therefore evaluates 20 fictional scenario families in which the same news supports the outcome, opposes it, or becomes irrelevant because the connecting relationship is removed. A fourth masked condition omits the relevant relationship and is not assigned a directional target.

Qwen2.5-72B, Llama-3.1-70B, and Qwen3-32B were evaluated with three repetitions per condition. Paired reversal requires the same news to produce correctly signed revisions in both the supporting and opposing contexts, while broken-link movement measures response when the stated connection to the outcome is absent. Paired reversal is 30.0–36.7% across the three models, and mean absolute movement in the broken-link condition is 10.5–13.3 percentage points. By contrast, no-news updates reproduce their baselines exactly, and repeated prior information produces mean absolute movement below 0.6 percentage points.

GPT-5.6 was evaluated once on the same 20 families. It gives the correct direction on 37 of 40 signed updates, succeeds on 17 of 20 reversal pairs, and moves by an average of 0.5 percentage points in the broken-link condition. This establishes that the task permits successful relational reversal under at least one tested deployment, but the comparison across models is descriptive because inference settings differ.

These results are limited by the small number of recurring mechanisms, the absence of independent human validation, endpoint constraints on some elicited baseline probabilities, and ambiguities in several scenario descriptions. The finite-rule context-reversal experiment in Section B.6 removes these weaknesses by deriving directional targets from explicit probabilistic rules and evaluating a larger set of controlled scenarios.

B.6 FINITE-RULE CONTEXT REVERSAL

The exploratory context-reversal study above leaves uncertainty about the intended relationships in its hand-authored scenarios. We therefore construct 80 new scenarios whose directional effects follow from explicit probabilistic rules. Ten scenarios instantiate each of eight mechanism classes: detector calibration, prerequisite timing, component quorums, procedural waivers, proportional brokerage,

directed reachability, bounded feedback, and dependence among stored reports. The task changes only the relationship between the actor named in a report and the outcome-generating mechanism, allowing identical news to increase, decrease, or leave unchanged the outcome probability.

Rule-based construction. Each scenario contains three independent channels with local latent variables, including a binary state X whose probability is revised from $1/4$ to $3/4$ or from $3/4$ to $1/4$ by the report. The project completes when a final inspection succeeds and either a reserve team succeeds or the ordinary channel activates without the cancellation channel activating. If a and b denote the activation probabilities of the ordinary and cancellation channels, then

$$P(\text{complete}) = \frac{4}{5} \left\{ \frac{1}{5} + \frac{4}{5}a(1-b) \right\}.$$

A third archive channel has no effect on the outcome and provides no information about the other channels. Assigning the reported actor to the ordinary, cancellation, or archive channel therefore makes the same report increase, decrease, or leave unchanged the outcome probability. All three contexts contain the same inventory of channel descriptions, and reports that raise and lower $P(X = 1)$ are balanced. Exact probabilities are computed independently by two implementations and remain between .16 and .84, so no target direction is mechanically blocked by an endpoint. The resulting labels are defined by the probabilistic rules rather than by model or human judgments. The materials have not undergone independent human validation.

Reasoning and wording conditions. Each scenario is rendered in four versions: the base wording, consistent entity renaming, a paraphrase, and an identical-text resample under a different sampling seed. Each version contains positive, negative, and irrelevant relational contexts with separate baseline forecasts. Independent update branches then supply new evidence, no new information, or repeated information already available at baseline.

Qwen3-32B is evaluated with thinking enabled and disabled on the same scenarios and matched request seeds. Both conditions use temperature 0.7, top- $p = 1$, unconstrained decoding, an 8,192-token output limit, and a 16,384-token context limit. Invalid, truncated, or missing responses count as failures in directional scores and are not resampled. Thinking disabled yields 3,840/3,840 valid probabilities, while thinking enabled yields 3,803/3,840.

Paired reversal requires both the positive and negative versions of a scenario to move in the correct direction. Zero movement counts as failure for a directional target. Broken-link movement measures absolute revision when the report is assigned to the outcome-irrelevant channel. Intervals use 2,000 bootstrap resamples of whole scenarios, preserving contexts, wording variants, and reasoning conditions.

Table 5: **Reasoning enables context-sensitive reversal under explicit probabilistic rules.** Signs count correctly directed updates out of 160 signed contexts per wording, pairs count successful reversals out of 80, and broken movement is mean absolute revision when the report is outcome-irrelevant. Brackets give descriptive 90% whole-scenario bootstrap intervals.

Thinking	Wording	Signs	Pairs [90% CI]	Broken $ \Delta $, pp	Observed
Off	Base	84/160	14/80 [10.0, 25.0]	4.2 [3.1, 5.4]	80/80
Off	Renamed	86/160	14/80 [11.2, 25.0]	2.7 [1.9, 3.5]	80/80
Off	Paraphrased	81/160	12/80 [8.8, 21.2]	2.1 [1.5, 2.9]	80/80
Off	Same-text resample	83/160	11/80 [7.5, 20.0]	3.3 [2.5, 4.2]	80/80
On	Base	157/160	77/80 [92.5, 98.8]	0.2 [0.0, 0.5]	80/80
On	Renamed	159/160	79/80 [96.2, 100.0]	0.0 [0.0, 0.0]	79/80
On	Paraphrased	154/160	74/80 [87.5, 97.5]	0.0 [0.0, 0.0]	79/80
On	Same-text resample	155/160	75/80 [88.8, 97.5]	0.1 [0.0, 0.2]	78/80

Under base wording, enabling thinking increases paired reversal by 78.8 percentage points, with a 90% interval of [71.2, 86.2]. The corresponding contrasts are 81.2 points under renaming, 77.5 under paraphrasing, and 80.0 under identical-text resampling. Omitting one mechanism class at a time gives base-wording contrasts from 75.7 to 82.9 points. The result is therefore not concentrated in one mechanism or wording variant.

The control branches distinguish relational updating from generic forecast instability. With thinking enabled, mean absolute movement under no new information is at most 0.06 percentage points

across wording variants, and repeated prior information produces at most 0.05 points of movement. Under base wording, exact-probability error falls from 28.75 percentage points at baseline with thinking disabled to 0.13 with thinking enabled, and updated-probability error falls from 26.89 to 0.08. The reasoning effect therefore extends beyond choosing the correct revision sign to recovering the rule-derived numerical probabilities.

Relation-invariant shortcuts. Within each scenario and wording, the positive, negative, and irrelevant contexts form a triple with the same news, named actor, and rule-derived baseline probability. A predictor that returns the same revision across the triple cannot satisfy all three targets. Because the scoring tolerance for an unchanged forecast overlaps an arbitrarily small positive revision, such a predictor can achieve at most 2/3 item-level directional accuracy but zero complete-triple accuracy. For exact posterior agreement within 10^{-4} , a constant posterior can match at most one of the three contexts.

Simple lexical classifiers likewise do not recover the target relation. When evaluation holds out complete scenarios and all of their wording variants, news-only, full-text unigram, and full-text bigram classifiers obtain approximately 1/3 directional accuracy and .50 balanced relevance accuracy. These controls do not rule out every possible shortcut, but they rule out predictors based only on the tested relation-invariant outputs or lexical features.

Matched frontier comparison. GPT-5.6 is evaluated on a balanced 24-scenario subset containing three scenarios from each mechanism class under base and paraphrased wording. On this same subset, GPT-5.6 produces the correct direction on all 48 signed updates and succeeds on all 24 reversal pairs under both wording variants, with zero mean broken-link movement. Qwen3-32B with thinking enabled succeeds on 22/24 base and 23/24 paraphrased reversal pairs, compared with 4/24 and 3/24 with thinking disabled. GPT-5.6 also matches all 144 rule-derived posteriors within 10^{-4} on this subset. Inference settings differ between GPT-5.6 and Qwen3-32B, so these results establish task solvability across deployments rather than an isolated model comparison.

Scope. The 80 scenarios instantiate eight recurring probabilistic mechanisms and therefore do not constitute 80 independent mechanism families or representative coverage of natural news. The materials share a common outcome template and have not undergone independent human validation. The experiment establishes sensitivity to explicitly specified relational structure in these controlled tasks rather than unrestricted causal reasoning in natural forecasting settings.

B.7 RELATIONAL DEPTH

A final diagnostic varies how directly the actor-to-channel relationship is expressed while preserving the same probabilistic calculation. Sixteen scenarios per condition express the assignment directly, through one intermediate relation, through two intermediate relations, or through two relations plus irrelevant actor-channel associations. Within every positive-negative-irrelevant triple, the news, named actor, and rule-derived baseline remain fixed while only the relational assignment changes.

Qwen3-32B with thinking enabled remains highly accurate across all four conditions.

Table 6: **Qwen3-32B under increasingly indirect relational descriptions.** Each condition contains 16 triples and 48 updates. Triple success requires both signed revisions to be correct and the irrelevant revision to remain within 10^{-4} of baseline. Exact counts forecasts within 10^{-4} of the rule-derived posterior.

Binding	Triples	Signs	Exact	Error (pp)
Direct statement	16/16	48/48	44/48	0.011
One step, via permits	15/16	46/48	44/48	0.007
Two steps, via permits and offices	16/16	48/48	45/48	0.050
Two steps, plus inert relations	14/16	46/48	41/48	0.320

Triple success is 16/16, 15/16, 16/16, and 14/16 across the four conditions, so performance does not decline monotonically with relational depth. Posterior error nevertheless rises to 0.320 percentage points in the condition containing two relational steps and additional irrelevant associations. The

sample is small and does not establish a general relationship between relational depth and forecasting difficulty.

C EXPERIMENT 2: CONTEXTUAL RELATIONSHIP SELECTION AND REVISION IN COIN CITY

Experiment 2 tests whether contextual information determines which response relationship a model applies and whether direct observations from the target setting revise that relationship. Models forecast a post-news election poll in City C using varying amounts of City C evidence and, in some conditions, observations from reference cities A and B. All prompt conditions reuse the same 250 seeded episodes, allowing within-episode comparisons under identical numerical evidence.

C.1 DESIGN AND DATA-GENERATING PROCESS

C.1.1 LATENT REGIMES AND OBSERVATIONS

Each episode crosses two binary factors: whether City A or City B is the strongly responsive reference city, and whether City C is strongly or weakly responsive. The four cells contain 63, 62, 62, and 63 episodes, respectively. For episode e and city c , the response coefficient is sampled independently and remains fixed across that city’s observations:

$$g_{ec} \sim \begin{cases} \mathcal{N}(0.90, 0.03^2), & \text{strong regime,} \\ \mathcal{N}(0.25, 0.03^2), & \text{weak regime.} \end{cases}$$

A regime cue therefore identifies the distribution of responsiveness without revealing the realized city-specific coefficient.

Each city contains four observed cases arranged as two matched pairs. Rows within a pair share a starting poll and news magnitude but have opposite news signs. For case r ,

$$p_{ecr}^{\text{end}} = \text{round}_{0.1} (p_{ecr}^{\text{start}} + g_{ec}x_{ecr} + \epsilon_{ecr}), \quad \epsilon_{ecr} \sim \mathcal{N}(0, 5^2).$$

Starting polls are sampled uniformly from $[35, 65]$. Reference-news magnitudes are drawn from $\{6, 7, 8, 9, 10\}$, and query shocks are drawn from $\{-10, \dots, -4, 4, \dots, 10\}$. For query starting poll p_0 and news value x , the scoring target is the latent conditional expectation $p_0 + g_C x$. No additional query-time observation noise is added to this target.

The evidence level $k \in \{0, 1, 2, 3, 4\}$ exposes the first k City C observations. Values $k = 2$ and $k = 4$ complete one and two matched pairs, while $k = 1$ and $k = 3$ expose partial pairs. Conditions containing reference-city evidence always show all four observations from Cities A and B.

Table 7: Coin City design.

Quantity	Value
Episodes	250
Target observations	$k \in \{0, 1, 2, 3, 4\}$
Primary conditions	target only; references without City C context; references with correct City C context
Additional conditions	misleading City C context; counterbalanced arbitrary labels
Strong / weak slope means	0.90/0.25 poll points per news point
Within-regime slope SD	0.03
Observation-noise SD	5 poll points
Reference cases per city	4, arranged as two matched pairs

C.2 PROMPT CONDITIONS AND AVAILABLE INFORMATION

All prompts state that rows are separate polling cases rather than a time series, that positive news favors the candidate, and that within-city responsiveness is stable despite noisy individual polls. Rows sharing a pair label are described as having the same starting poll and equal-magnitude, opposite-sign

news. Every displayed row gives the starting poll, news value, poll change, and ending poll. The required answer contains a numeric `predicted_poll` between 0 and 100 and a one-sentence rationale.

C.2.1 FIVE INFORMATION CONDITIONS

Target only. The model sees the first k City C observations and no reference-city evidence or contextual description.

References, no City C context. The model additionally sees all four observations from Cities A and B. The strongly responsive reference is described as receiving national news coverage, where campaign stories tend to produce larger immediate poll movements. The weakly responsive reference is described as receiving local news coverage, where campaign stories tend to produce smaller immediate poll movements. City C receives no contextual description identifying which reference regime applies.

References plus correct City C context. The numerical evidence is identical to the preceding condition. The only addition is a sentence describing City C as receiving national or local coverage, correctly aligned with its latent response regime.

References plus misleading City C context. The prompt is identical to the correct-context condition except that the City C sentence names the opposite response regime. All numerical observations and forecast queries remain unchanged.

References plus arbitrary City C label. The national/local descriptions are replaced by the arbitrary labels `KIV` and `ZOR`. The prompt states only that cities sharing a label share a typical response regime and that the labels have no inherent meaning or ordering. The mapping between label and response regime reverses across episodes. At $k = 0$, the meaning of City C’s label can therefore be recovered only from the labeled numerical examples from Cities A and B.

No prompt contains a fitted coefficient, the strong or weak numerical means, the observation-noise scale, an estimator, or a rule for pooling evidence across cities. The correct-context versus no-context contrast changes only the City C description while holding all displayed numerical evidence fixed. The correct-context versus misleading-context contrast changes only the regime named by that description. The arbitrary-label versus no-context contrast tests whether an episode-specific relationship can be induced without relying on the meanings of *national* and *local*.

Semantic cues versus induced relationships. The semantic descriptions explicitly state which media environment tends to produce larger or smaller responses. Success in the semantic conditions therefore demonstrates numerical use of a supplied qualitative relationship rather than induction of that ordering from observations. For example, a fixed policy that ignores the displayed cases and assigns coefficients 1.0 and 0.2 to the national and local regimes obtains $k = 0$ forecast MAE 0.523 and implied-slope correlation .996. The semantic condition alone therefore cannot establish that the model inferred the relationship from the displayed reference cases. The stronger induction tests are the arbitrary-label condition and the reference-selection analysis in Section C.6, where the mapping must be recovered from episode-specific examples.

C.3 MODEL EXECUTION

C.3.1 PRIMARY DEPLOYMENTS

The primary evaluation includes Claude Opus 4.8, GPT-5.6, DeepSeek V4-Pro, Kimi K3, Gemini 3.6 Flash, and Claude Opus 5. No deployment uses tools or retrieval. GPT-5.6 and Claude Opus 5 receive no explicit temperature parameter, while the remaining deployments use temperature zero. The six deployments differ in provider, reasoning mode, and output budget, so cross-model comparisons are descriptive rather than configuration-matched capability comparisons.

Table 8: **Primary deployment settings and usable responses.** Parsed counts are usable numerical forecasts out of 3,750 primary tasks per model.

Model	Reasoning / sampling	Token limit	Parsed
Claude Opus 4.8	temperature 0	512	3,750
GPT-5.6	low reasoning; temperature omitted	4,096	3,749
DeepSeek V4-Pro	temperature 0	4,096	3,750
Kimi K3	low reasoning; temperature 0	16,384	3,750
Gemini 3.6 Flash	low reasoning; temperature 0	16,384	3,750
Claude Opus 5	default extended thinking; temperature omitted	4,096	3,748

A response is usable when `predicted_poll` parses to a finite number in $[0, 100]$. Primary comparisons use complete cases across the compared conditions, and unsuccessful responses remain missing rather than being re-elicited. Section C.9 reports the resulting missingness.

C.3.2 MATCHED MISLEADING-CONTEXT INTERVENTION

Each misleading-context prompt differs from its correct-context counterpart by exactly one City C sentence. The City C description names the opposite response regime, while all numerical observations, queries, and other instructions remain unchanged. Across the 1,250 episode–evidence combinations, 625 strong-response targets are described as weak and 625 weak-response targets as strong. Each deployment uses the same inference settings in the correct and misleading conditions.

C.3.3 ARBITRARY-LABEL INTERVENTION

The arbitrary-label condition reuses the same numerical task content while replacing semantic regime descriptions with KIV and ZOR. The more responsive regime is labeled KIV in 125 episodes and ZOR in 125 episodes, giving 62 or 63 episodes in each target-regime–label combination. At $k = 0$, City C contributes no observations, so identifying its regime requires inferring the label mapping from the reference cities.

The arbitrary-label comparison includes Qwen3 4B/8B/14B/32B, Qwen2.5 7B/14B/32B/72B, Llama-3.1-8B/70B, DeepSeek V4-Pro, Kimi K3, Claude Opus 4.8, Claude Opus 5, and GPT-5.6. Each model is evaluated under no City C context, semantic City C context, and arbitrary-label context on the same 1,250 episode–evidence tasks. Open-weight checkpoints run under vLLM in bfloat16 with deterministic decoding; Qwen uses its no-thinking chat template and Llama uses its native instruction template.

For DeepSeek V4-Pro, the three arbitrary-label comparison conditions use the same deployment and inference configuration. Kimi K3, Claude Opus 5, and GPT-5.6 use their corresponding primary configurations. Claude Opus 4.8’s arbitrary-label condition ran on a second Azure resource serving the same model and used a 1,024-token rather than 512-token output limit. Its within-model arbitrary-label contrast is therefore retained as a robustness result, but provider-level comparisons involving its magnitude should be interpreted cautiously.

Arbitrary-label analyses use episodes with usable forecasts in all three compared conditions. Unsuccessful or unresolved responses remain missing.

C.4 ESTIMANDS, BENCHMARKS, AND STATISTICAL INFERENCE

C.4.1 FORECAST ERROR AND INFERRED RESPONSIVENESS

For model forecast $\hat{p}_e$, query starting poll p_{0e} , news value x_e , and latent City C response coefficient g_{Ce} , forecast error is

$$\text{MAE} = \frac{1}{n} \sum_{e=1}^n |\hat{p}_e - (p_{0e} + g_{Ce}x_e)|.$$

The target is the latent conditional expectation and contains no additional query-time observation noise.

Because every query has $x_e \neq 0$, each forecast also implies a response coefficient

$$\hat{g}_e = \frac{\hat{p}_e - p_{0e}}{x_e}.$$

We report mean absolute error in $\hat{g}_e$ and its Pearson correlation with g_{Ce} . Slope correlation measures whether forecasts distinguish more- from less-responsive cities, while slope MAE measures numerical accuracy of the inferred response magnitude. A nearly constant response coefficient can therefore achieve moderate forecast MAE while having little correlation with the true episode-specific coefficient.

Within each model and evidence level, primary arm comparisons use the common set of episodes with usable forecasts in all compared conditions. Arbitrary-label contrasts likewise use matched complete cases across no-context, semantic-context, and arbitrary-label conditions. All contrasts are paired by episode.

C.4.2 REFERENCE-SELECTION BENCHMARK

The label-conditioned analogical benchmark fits separate through-origin response coefficients for Cities A and B and selects the reference whose stated regime matches City C. At $k = 0$, it transfers that reference coefficient directly. At $k > 0$, it pools the selected reference observations with the available City C observations before refitting. Under a misleading cue, the same procedure selects the opposite reference. At $k = 0$, this benchmark obtains forecast MAE 1.842 and slope correlation .656 under the correct cue, compared with MAE 4.561 and correlation $-.667$ under the misleading cue. The benchmark demonstrates that a simple reference-selection policy can reproduce the qualitative behavioral pattern without implying that models implement this particular estimator.

C.4.3 UNCERTAINTY

Paired condition contrasts use 90% percentile intervals (5th and 95th percentiles) from episode-level bootstrap resampling. Unless otherwise stated, intervals use 2,000 bootstrap draws and preserve pairing between conditions within each episode. Intervals are pointwise rather than simultaneous across models, evidence levels, or metrics, and no multiplicity correction is applied.

C.5 BEHAVIORAL RESULTS

C.5.1 CONTEXT SELECTS THE ZERO-SHOT RESPONSE RELATIONSHIP

With no City C observations, target-only and no-context forecasts show little relationship to City C’s latent response coefficient. Adding the correct City C description raises slope correlation to .523–.700 across all six primary deployments and lowers forecast MAE relative to the numerically identical no-context condition. Changing only that description to the opposite regime instead produces negative slope correlations between $-.50$ and $-.68$.

Table 9: Zero-shot effect of City C context. All three conditions use the same reference-city observations and forecast query at $k = 0$; only the City C description changes between the no-context, correct-context, and misleading-context conditions. MAE is in poll points, and ρ correlates forecast-implied and true City C response coefficients.

Model	No context		Correct context		Misleading context	
	MAE	ρ	MAE	ρ	MAE	ρ
Claude Opus 4.8	2.651	-.063	1.817	.700	4.20	-.66
GPT-5.6	2.518	-.041	1.596	.678	4.23	-.68
DeepSeek V4-Pro	2.841	-.027	2.474	.523	4.36	-.62
Kimi K3	2.548	-.021	1.886	.593	4.31	-.62
Gemini 3.6 Flash	2.552	.078	2.174	.559	4.29	-.50
Claude Opus 5	2.619	-.007	1.764	.617	4.25	-.62

The misleading cue continues to produce negative slope correlation for all six deployments after one City C observation. After two observations, all six correlations become positive, and by $k = 4$ they

range from .32 to .47. Direct target evidence therefore increasingly overrides an initially incorrect contextual assignment.

Correct context does not uniformly minimize forecast error at every later evidence level. The value of the cue is strongest when City C evidence is sparse, while direct observations increasingly support response estimation without relying on the contextual assignment. This distinction is expected because relationship identification and numerical forecast accuracy are separate quantities.

C.5.2 ARBITRARY LABELS REQUIRE EPISODE-SPECIFIC INDUCTION

The semantic descriptions provide a pre-existing interpretation of national versus local coverage. The arbitrary-label condition removes this shortcut by assigning KIV and ZOR opposite meanings across episodes. At $k = 0$, City C contributes no observations, so its response regime can be identified only by inferring the episode-specific label mapping from Cities A and B.

Table 10: **Zero-shot arbitrary-label effects across 15 models.** Each contrast is arbitrary-label minus no-context at $k = 0$, using matched episodes. Negative ΔMAE indicates lower forecast error, while positive $\Delta\rho$ and $\Delta\text{accuracy}$ indicate better recovery of City C’s response regime. Brackets give pointwise 90% episode-bootstrap intervals.

Model	n	ΔMAE [90% CI]	$\Delta\rho$ [90% CI]	$\Delta\text{accuracy}$ [90% CI]
Qwen3-4B	250	−.005 [−.074, .063]	+ .095 [−.037, .215]	+ .016 [−.004, .040]
Qwen2.5-7B	250	+ .338 [.071, .616]	+ .264 [.137, .385]	+ .072 [.024, .116]
Qwen3-8B	250	−.011 [−.197, .175]	+ .023 [−.105, .146]	+ .024 [−.008, .056]
Qwen2.5-14B	250	+ .105 [−.134, .345]	+ .236 [.128, .336]	+ .088 [.032, .144]
Qwen3-14B	250	+ .180 [−.122, .492]	+ .398 [.276, .520]	+ .164 [.100, .228]
Qwen3-32B	250	+ .432 [.169, .696]	+ .302 [.201, .410]	+ .132 [.072, .184]
Qwen2.5-32B	250	+ .529 [.224, .842]	+ .372 [.255, .488]	+ .148 [.100, .200]
Qwen2.5-72B	250	+ .924 [.624, 1.241]	+ .381 [.269, .488]	+ .116 [.068, .164]
Llama-3.1-8B	242	+ .492 [.094, .903]	+ .045 [−.097, .216]	+ .045 [−.017, .107]
Llama-3.1-70B	250	+ .448 [.096, .802]	+ .424 [.330, .525]	+ .092 [.044, .140]
DeepSeek V4-Pro	250	+ .107 [−.171, .408]	+ .436 [.343, .535]	+ .172 [.112, .232]
Kimi K3	250	−.497 [−.732, −.262]	+ .633 [.539, .728]	+ .336 [.284, .392]
Claude Opus 4.8	250	−.584 [−.823, −.359]	+ .623 [.515, .729]	+ .312 [.252, .372]
Claude Opus 5	250	−.603 [−.819, −.385]	+ .591 [.499, .681]	+ .284 [.232, .340]
GPT-5.6	250	−.774 [−.980, −.561]	+ .696 [.614, .788]	+ .372 [.320, .424]

Twelve of the fifteen models have positive symbol-minus-no-context slope-correlation intervals that exclude zero. The three exceptions are Qwen3-4B, Qwen3-8B, and Llama-3.1-8B. Within Qwen3, the effect is detectable at 14B and 32B but not at 4B or 8B, while all four tested Qwen2.5 checkpoints show positive effects. Within Llama-3.1, the 70B checkpoint shows a positive effect while the 8B checkpoint does not. These patterns do not define a universal parameter threshold because capability varies across model families and is not monotonic within every family.

Recovering the label mapping does not guarantee lower forecast error. Several open-weight models improve response-regime discrimination while their forecast MAE increases, whereas Kimi K3, both Claude Opus deployments, and GPT-5.6 improve both discrimination and zero-shot forecast accuracy. DeepSeek V4-Pro improves discrimination without a clear MAE change. Relationship selection and numerical calibration are therefore distinct outcomes.

The arbitrary-label advantage also diminishes as direct City C observations accumulate. For Qwen2.5-32B and Qwen2.5-72B, the symbol-minus-no-context slope-correlation intervals span zero by $k = 3$ and $k = 4$. Llama-3.1-70B shows the same pattern at those evidence levels. The induced convention is therefore most useful when target-specific evidence is absent or sparse.

C.6 DOES THE FORECAST TRACK THE CUE-SELECTED REFERENCE?

Regime-level forecast correlations cannot distinguish a fixed semantic prior from numerical use of the displayed reference cities. A model could map *national* and *local* directly to different response magnitudes without using City A or City B, or it could use the cue to select a reference city and transfer information from that city’s observed response.

We distinguish these accounts using reference-specific variation within regimes. Let $\hat{\beta}_e^{\text{ref}}$ denote the through-origin slope fitted to a reference city’s four noisy observations, and let R_e denote the regime

named for City C. We measure

$$r(\hat{g}_e, \hat{\beta}_e^{\text{ref}} | R_e),$$

the Pearson correlation between the residuals obtained after regressing the forecast-implied slope and reference fitted slope separately on an intercept and the binary regime indicator. This removes the strong between-regime association and tests whether forecasts track reference-specific numerical variation beyond regime membership. Pointwise 90% intervals use 2,000 episode-bootstrap resamples.

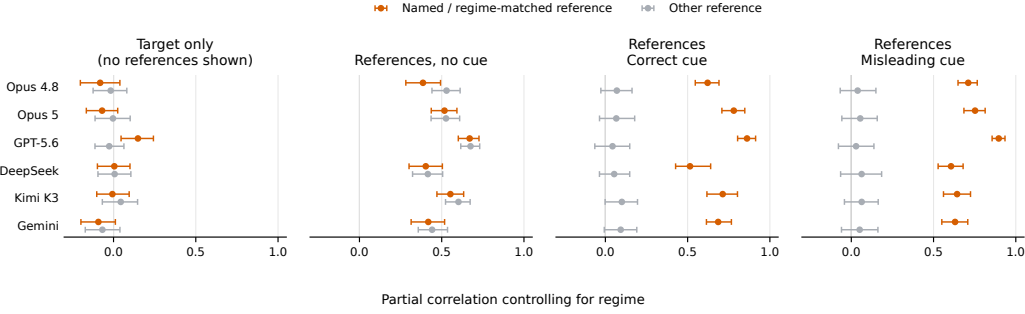


Figure 9: **Forecasts track the reference selected by the cue.** Points show $k = 0$ partial correlations between forecast-implied responsiveness and each reference city’s fitted responsiveness after controlling for regime. Whiskers give pointwise 90% episode-bootstrap intervals. Without a City C cue, forecasts track both displayed references. Correct and misleading cues concentrate the association on the reference selected by the cue.

Without reference-city observations, the corresponding associations are near zero. When both references are displayed but City C receives no cue, all six primary deployments are positively associated with both reference estimates. A correct City C cue concentrates this association on the regime-matched reference, with all six named-reference intervals excluding zero and all six other-reference intervals spanning zero. A misleading cue produces the same pattern for the incorrectly selected reference. Forecasts therefore depend on reference-specific numerical variation that cannot be explained by a response determined solely by the semantic cue.

The same test under arbitrary labels gives concordant results. Twelve of fifteen models have positive symbol-matched reference correlations whose 90% intervals exclude zero. The three exceptions are Qwen3-4B, Qwen3-8B, and Llama-3.1-8B, which are also the three models whose behavioral symbol-minus-no-context slope-correlation intervals span zero. Because the behavioral and reference-selection analyses use the same forecasts, this agreement is complementary evidence rather than an independent replication.

The partial correlation measures association with a reference estimate rather than the weight assigned to that estimate. Shrinking a noisy reference slope toward a regime-level estimate can preserve its residual correlation, so positive values do not imply inefficient transfer of reference noise. Conversely, a null association does not establish reliance on a lexical prior. The analysis therefore supports numerical reference selection without identifying the model’s internal estimator.

C.7 REPRESENTATION AND CAUSAL ANALYSES

The behavioral analyses establish that some models infer and use an episode-specific reference mapping, but they do not identify how that mapping is represented internally. We therefore test whether the target regime is linearly decodable from model activations and whether intervening on label-position states changes the resulting forecast.

C.7.1 LINEAR-PROBE METHODOLOGY

For each prompt, we extract the hidden state at the final input token from the embedding layer and every transformer block. Separate ridge probes are fit for each layer, target, and evidence depth using development episodes only. Features are standardized on the fitting data, and four-fold cross-

validation selects both the ridge penalty and probe layer before evaluation on held-out episodes. No test performance is used for layer or regularization selection.

The probe targets are the binary strong-versus-weak regime, the realized response coefficient g_C , and the within-regime residual

$$g_C - \mu_{R_e},$$

where $\mu_{R_e} = .90$ for the strong regime and $.25$ for the weak regime. Regime decoding is scored by ROC AUC, while continuous targets use R^2 . Mean-pooled input embeddings provide a control for information already linearly available from the prompt before transformer processing.

One-sided permutation tests evaluate held-out probe performance with 10,000 target-label permutations while keeping predictions, fitted weights, selected layer, and regularization fixed. The tests are reported individually without correction across models, targets, or evidence depths.

Why arbitrary labels are the informative probe condition. Semantic descriptions explicitly identify City C as belonging to the national or local regime. Accordingly, the semantic regime is highly linearly decodable even from input embeddings, and high semantic-cue probe performance does not by itself demonstrate an induced internal relationship. The arbitrary-label condition provides the stronger test because the meanings of KIV and ZOR reverse across episodes and must be inferred from the displayed reference examples. The larger-sample arbitrary-label probes and causal interventions below therefore carry the primary representational claim.

C.7.2 PROBE TARGETS AND INFORMATION LIMITS

The probe analysis distinguishes regime identification from recovery of City C’s exact response coefficient. Under the data-generating process, knowing the true regime mean explains approximately .992 of the variance in g_C , because the within-regime standard deviation is only .03. By contrast, City C’s within-regime residual is independent of the reference observations at $k = 0$, giving population $R^2 = 0$, and four City C observations provide only a small amount of residual information, with an approximate benchmark of $R^2 = .009$. High full-slope R^2 can therefore arise almost entirely from identifying the response regime. We consequently treat regime decoding as the primary representational endpoint and use the residual target only as a specificity check.

Under arbitrary labels, comparing the fitted slopes of the two reference cities identifies the target regime with simulated AUC .979. This value describes one estimator available from the displayed examples rather than an optimal ceiling over all possible estimators.

C.7.3 ARBITRARY-LABEL REGIME DECODING

The larger probe evaluation uses 248 episodes, with 160 development episodes and 88 held-out test episodes. Each episode is rendered at $k = 0$ and $k = 4$ under correct, absent, and misleading arbitrary-label assignments. All seven arbitrary-label models use the same train–test split. Probe layers and ridge penalties are selected using development episodes only, and the held-out episodes are not used for fitting or model selection.

At $k = 0$, the input embeddings themselves provide little linear information about the arbitrary mapping, with mean-pooled embedding AUCs between .495 and .556 across the seven models. After transformer processing, four models show significant held-out regime decoding: Qwen3-14B, Qwen2.5-32B, Qwen2.5-72B, and Llama-3.1-70B. The remaining three models, Qwen3-4B, Qwen3-8B, and Llama-3.1-8B, do not show significant $k = 0$ regime decoding.

Table 11: **Zero-shot decoding of an induced arbitrary-label regime.** Probe AUC is held-out regime decoding at $k = 0$; Embed. is the corresponding mean-pooled input-embedding AUC. The p -value is from a one-sided 10,000-permutation test. Behavioral $\Delta\rho$ is the symbol-minus-no-context slope-correlation effect from Table 10.

Model	Probe AUC	Embed. AUC	p	Behavioral $\Delta\rho$
Qwen3-4B	.520	.498	.376	.095
Qwen3-8B	.566	.534	.147	.023
Qwen3-14B	.678	.555	.0016	.398
Qwen2.5-32B	.778	.550	< .001	.372
Qwen2.5-72B	.662	.556	.0026	.381
Llama-3.1-8B	.423	.539	.893	.045
Llama-3.1-70B	.728	.495	< .001	.424

The same four models with significant $k = 0$ regime probes also have behavioral symbol-minus-no-context $\Delta\rho$ intervals excluding zero. The other three models have behavioral intervals spanning zero. This cross-check links linear decodability with behavioral use of the induced convention across checkpoints, although the two analyses use related tasks and are not independent replications.

Qwen3-14B illustrates how the representation changes when direct City C evidence becomes available. At $k = 0$, its arbitrary-label probe has AUC .678 under the correct label, compared with .467 with no label and .321 under the misleading label. At $k = 4$, the misleading-arm probe instead favors the true regime, with AUC .747, indicating that direct target observations can displace the initially supplied mapping. This does not imply that all influence of the label disappears.

The checkpoint pattern should not be interpreted as a universal scale threshold. Within Qwen3, the $k = 0$ effect is detected at 14B but not at 4B or 8B, while Llama-3.1 differs between 8B and 70B. Model size also changes architecture and training, and the robustness population in Section C.8 changes which individual Qwen3 intervals exclude zero.

C.7.4 CAUSAL INTERVENTION ON LABEL-POSITION STATES

Linear decoding does not establish that the decoded information influences the forecast. We therefore intervene on the hidden states at the two City C label-token positions under arbitrary labels. Correct-label and swapped-label prompts are token-aligned and differ only in City C’s label. For each recipient prompt, the intervention replaces the label-position states with the corresponding states from the opposite-label donor and then generates the forecast greedily.

For episode e , let $s_e = +1$ for a strong-response City C and $s_e = -1$ for a weak-response City C. The symmetric intervention effect is

$$d_e = \frac{s_e}{2} [(\hat{g}_{W \leftarrow C, e} - \hat{g}_{W, e}) + (\hat{g}_{C, e} - \hat{g}_{C \leftarrow W, e})],$$

where C and W denote the correct- and swapped-label prompts and the arrow denotes transfer from donor to recipient. Positive d_e means that replacing the recipient state moves its forecast toward the donor label’s response regime.

An initial held-out analysis of Qwen3-14B selected transformer outputs 19–21 using probe performance on 160 development episodes and obtained a $k = 0$ mean shift of .333 with a 90% bootstrap interval of [.198, .471] on 88 test episodes. The selected window produced no positive average shift at $k = 4$. Because intervention-window selection can itself introduce researcher degrees of freedom, the primary evidence below uses a fresh confirmation set with the window fixed before any new episodes are generated.

C.7.5 FIXED-WINDOW CONFIRMATION ON FRESH EPISODES

The confirmation uses 88 new episodes from the unchanged generator, balanced across reference-city regime, City C regime, and arbitrary label. None overlaps the development or test episodes used above. Before evaluating these episodes, the intervention windows, hypotheses, statistical tests, multiplicity correction, and validity criteria were fixed.

Both Qwen3-14B and Llama-3.1-70B are evaluated at transformer outputs 19–21 and at a later comparison window. For Qwen3-14B, outputs 19–21 were selected on the preceding 160-episode

development set. For Llama-3.1-70B, outputs 19–21 were chosen from a partial development sweep as a representative location on a broad intermediate-depth plateau rather than as the maximum observed effect. The late comparison windows are outputs 33–35 for Qwen3-14B and 35–37 for Llama-3.1-70B.

The primary hypotheses are that the mean d_e at outputs 19–21 is positive at $k = 0$ and that this effect exceeds the corresponding late-window effect. Tests use one-sided episode-level sign flips with 200,000 draws, with the four primary tests Holm-corrected at familywise level .05. Intervals use 10,000 episode-bootstrap resamples. Pre-specified validity gates require at least 77 of 88 complete episodes and at least 319 of 352 self-patched forecasts to reproduce their unpatched counterparts exactly. Both models satisfy both gates.

Table 12: **Fixed-window intervention on 88 fresh episodes.** Shift is the mean symmetric effect d_e in slope units, with 90% episode-bootstrap intervals. Positive values indicate movement toward the donor label’s regime. The late window is outputs 33–35 for Qwen3-14B and 35–37 for Llama-3.1-70B.

Patch site	k	Qwen3-14B		Llama-3.1-70B	
		Shift [90% CI]	p	Shift [90% CI]	p
Outputs 19–21	0	.452 [.343, .562]	$< 5 \times 10^{-6}$	.641 [.503, .778]	$< 5 \times 10^{-6}$
Outputs 19–21	4	-.016 [-.094, .059]	.634	.074 [.021, .129]	.012
Late window	0	.044 [.010, .082]	.022	-.013 [-.026, -.001]	.955
19–21 minus late	0	.376 [.280, .475]	$< 5 \times 10^{-6}$	.654 [.516, .792]	$< 5 \times 10^{-6}$

Both pre-specified $k = 0$ hypotheses are supported for both models after Holm correction. Patching outputs 19–21 shifts the implied response coefficient toward the donor label’s regime by .452 in Qwen3-14B and .641 in Llama-3.1-70B. The intermediate-minus-late contrasts are .376 and .654, respectively, showing that the effect is substantially larger at the selected intermediate depth than at the late comparison window.

Replacing the input embeddings at the label positions produces $k = 0$ shifts of .528 for Qwen3-14B and .665 for Llama-3.1-70B. Thus, changing only three intermediate label-position states transfers most of the effect produced by replacing the label representation at the input.

With four City C observations, the Qwen3-14B intermediate-window effect is indistinguishable from zero. Llama-3.1-70B retains a smaller positive effect of .074, indicating that direct target evidence reduces but does not uniformly eliminate reliance on the label. The experiment therefore supports a causal role for label-position states in zero-shot regime selection while also showing that their influence changes as direct evidence accumulates.

C.7.6 INTERPRETATION AND SCOPE

The arbitrary-label probes establish linear decodability of an episode-specific mapping in four of seven tested open-weight checkpoints. The causal interventions go further by showing that replacing label-position states changes zero-shot forecasts in Qwen3-14B and Llama-3.1-70B on fresh episodes.

These interventions replace complete hidden-state vectors rather than a single decoded direction. They therefore do not establish that the linear probe identifies the unique causal representation, that one representation fully mediates the behavioral effect, or that the same mechanism operates in hosted models. The causal evidence is restricted to two checkpoints from two model families and one intermediate-depth intervention.

Finally, the task asks models to select between two well-separated response regimes. It does not test continuous contextual variables or fine-grained recovery of City C’s within-regime deviation. Regime identification, reference selection, and exact coefficient estimation remain distinct capabilities.

C.8 ROBUSTNESS ACROSS GENERATOR POPULATIONS AND DEPLOYMENTS

C.8.1 HARDER GENERATOR POPULATION

We repeat the arbitrary-label evaluation on 250 new episodes with strong and weak regime means of .75 and .40 and within-regime slope standard deviation .10. This jointly reduces between-regime separation and increases within-regime heterogeneity relative to the original population. Observation noise, factorial balance, prompts, evidence depths, checkpoint settings, and episode-randomized symbol mappings are otherwise unchanged. Ten open-weight models are evaluated under no City C context, semantic context, and arbitrary-label context at all five evidence depths.

Table 13: **Zero-shot arbitrary-label effects across generator populations.** Values are symbol-minus-no-context changes in slope correlation at $k = 0$, with pointwise 90% episode-bootstrap intervals. The harder population narrows regime separation and increases within-regime dispersion.

Model	Original $\Delta\rho$ [90% CI]	Harder $\Delta\rho$ [90% CI]
Qwen3-4B	+.095 [−.037, .215]	+.050 [−.058, .178]
Qwen3-8B	+.023 [−.105, .146]	+.231 [.101, .345]
Qwen3-14B	+.398 [.276, .520]	+.155 [.023, .281]
Qwen3-32B	+.302 [.201, .410]	+.193 [.085, .301]
Qwen2.5-7B	+.264 [.137, .385]	+.149 [.030, .259]
Qwen2.5-14B	+.236 [.128, .336]	+.133 [.026, .233]
Qwen2.5-32B	+.372 [.255, .488]	+.247 [.131, .361]
Qwen2.5-72B	+.381 [.269, .488]	+.336 [.226, .444]
Llama-3.1-8B	+.045 [−.097, .216]	−.077 [−.193, .056]
Llama-3.1-70B	+.424 [.330, .525]	+.242 [.139, .347]

Seven of ten models have positive intervals excluding zero in both populations, and nine of ten retain the same interval classification. All four Qwen2.5 checkpoints remain positive in the harder population, while Llama-3.1-8B spans zero in both populations and Llama-3.1-70B is positive in both. Within Qwen3, the checkpoints whose intervals exclude zero change across populations. The results therefore support robustness of arbitrary-label induction to this joint generator perturbation without implying a universal capability threshold in parameter count.

Forecast discrimination and numerical accuracy remain distinct. For example, Qwen2.5-72B improves slope discrimination in the harder population while arbitrary-label context increases its forecast MAE by 1.012 poll points with a 90% interval of [.760, 1.271]. The harder population changes regime separation and within-regime dispersion simultaneously, so it does not identify the effect of either factor in isolation.

C.9 RESPONSE VALIDITY AND SCOPE

C.9.1 MISSING FORECASTS

The primary evaluation contains 22,500 model–prompt combinations, of which three lack a usable forecast. Matched primary comparisons therefore use $n = 249$ for the affected model–prefix combinations and $n = 250$ elsewhere. All 7,500 misleading-context tasks yield usable analyzed forecasts after deduplication.

The 15-model arbitrary-label comparison contains 56,250 model–prompt combinations and uses complete cases across the three compared conditions. At $k = 0$, every model contributes 250 complete episodes except Llama-3.1-8B, which contributes 242. Across later evidence levels, its complete-case sample ranges from 238 to 246. Unusable forecasts remain missing rather than being imputed.

C.9.2 INTERPRETATION AND SCOPE

Experiment 2 provides convergent evidence for context-sensitive relationship selection. Matched cue interventions show that changing only City C’s contextual assignment changes the inferred response relationship. Arbitrary labels show that this assignment can be induced from episode-specific examples rather than supplied by familiar semantics. Reference-selection analyses show that forecasts track numerical variation in the reference selected by the cue. Linear probes show

that the induced arbitrary-label regime is decodable in several open-weight checkpoints, and the fixed-window interventions show that label-position states causally influence zero-shot forecasts in Qwen3-14B and Llama-3.1-70B.

These analyses answer different questions and are not statistically independent replications. Behavioral contrasts measure forecast changes, reference-selection correlations measure association with displayed numerical examples, probes measure accessibility to a linear readout, and patching measures the causal effect of replacing selected hidden states. None identifies a unique internal algorithm.

The scoring target is City C’s noise-free conditional expectation rather than a future noisy poll realization. The prompts explicitly state within-city stability and the matched-pair structure, and the contextual variable is binary and well separated. The experiment therefore does not establish performance under ambiguous, continuous, or adversarial contextual relationships.

Cross-model comparisons are descriptive because deployments differ in provider, reasoning configuration, and output budget. The two generator populations jointly vary regime separation and within-regime heterogeneity. Except where explicitly corrected in the fixed-window causal confirmation, reported bootstrap and permutation intervals are pointwise and do not provide simultaneous coverage across the many models, metrics, and evidence levels.

D EXPERIMENT 3: CONTROLLED RELATIONSHIP TRANSFER

Experiment 3 asks whether outcome-based training improves use of an episode-specific response relationship when the forecasting domain or response mechanism changes. Training uses direct-response systems in Coin City, while evaluation independently varies the surface domain, response mechanism, and amount of target-specific evidence. The primary comparison holds training prompts and the distribution of reward targets fixed while changing whether each prompt is paired with its own episode-specific target. Qwen3-4B and Qwen3-8B are each evaluated with eight training seeds per reward arm.

D.1 DESIGN AND EVALUATION

D.1.1 DOMAINS AND RESPONSE MECHANISMS

Let x_t denote the input and z_t the latent deviation of the outcome from its resting level. Training uses the direct-response mechanism,

$$z_{t+1} = g_j x_t. \quad (7)$$

A pulse therefore affects the next observation but does not persist after subsequent inputs return to zero.

Evaluation additionally includes a mediated mechanism,

$$m_{t+1} = \rho m_t + g_j x_t, \quad (8)$$

$$z_{t+1} = \phi z_t + \lambda m_{t+1}. \quad (9)$$

Both latent states begin at zero. Observed outcomes add independent Gaussian noise and are clipped to the stated domain range, while evaluation targets use the corresponding noise-free trajectory. Prompts provide the domain bounds and qualitative pathway descriptions but omit the equations, response parameters, mediator, and structure labels.

Coin City has resting level 50, support $[0, 100]$, and observation-noise standard deviation 1.6. Coin Harbor changes the terminology, units, numerical scale, input magnitudes, and pathway descriptions, with resting level 180, support $[40, 320]$, and observation-noise standard deviation 4.5. Training reference gains are sampled from $[\text{.18}, \text{.46}]$ for the weak-response system and $[\text{.68}, \text{.98}]$ for the strong-response system. At evaluation, $g \sim \text{Uniform}(\text{.22}, \text{.92})$, while Structure B uses $\rho \sim \text{Uniform}(\text{.48}, \text{.74})$, $\phi \sim \text{Uniform}(\text{.18}, \text{.42})$, and $\lambda \sim \text{Uniform}(\text{.68}, \text{.94})$. Coin City inputs are drawn from $\{-12, -8, -4, 0, 4, 8, 12\}$, while Coin Harbor uses $\{-24, -16, -8, 0, 8, 16, 24\}$. Evaluation queries use shocks $\{-12, -6, 0, 6, 12\}$ in Coin City and $\{-24, -12, 0, 12, 24\}$ in Coin Harbor at horizons one and three. A reference and its matched target share these latent response parameters but have independent inputs and observation noise.

Table 14: **Training and transfer conditions.** Training uses Coin City with the direct response mechanism, while evaluation independently changes the surface domain and response structure.

Evaluation cell	Domain	Response structure
In distribution	Coin City	Direct, Structure A
Domain shift	Coin Harbor	Direct, Structure A
Structure shift	Coin City	Mediated, Structure B
Joint shift	Coin Harbor	Mediated, Structure B

D.1.2 EPISODE-MATCHED TRAINING AND POPULATION-PRIOR CONTROL

Each training prompt contains two complete 14-case reference systems with different response gains and a target that shares one reference’s gain. The target reveals $k \in \{0, 2, 4, 8\}$ observations, and its qualitative background identifies the matching response regime. Training contains 1,200 prompts at each evidence depth, for 4,800 prompts in total. Weak- and strong-response targets are balanced overall but associated with evidence depth, so k itself provides some information about the gain band.

The episode-matched and population-prior conditions receive identical prompts and the same multiset of answer vectors. Episode-matched reward scores a response against the forecasts generated by that

episode’s own latent parameters. The population-prior control reassigns complete answer vectors among episodes with the same evidence depth so that every vector is used exactly once and no episode retains its own vector. This preserves the reward distribution conditional on k while removing the episode-specific prompt–target pairing. The contrast therefore tests episode-specific supervision beyond learning the answer distribution associated with evidence depth and other shared prompt features.

Training reward and output contract. The Experiment 3 reward is

$$r = .4r_{\text{level}} + .6r_{\text{response}},$$

where each component averages $\max(0, 1 - |e|/S)$ over its corresponding forecast errors. The level tolerance is 10% of the domain’s observable range, while the response tolerance is half one intervention unit. Training and evaluation require exactly ten finite forecasts in a JSON object; malformed or length-truncated outputs receive zero training reward.

D.1.3 EVALUATION PROTOCOL

Every evaluation prompt contains one complete direct-response reference, one complete mediated-response reference, and a target generated by one of those mechanisms. The target receives either the correct qualitative pathway description, no pathway description, or the description of the other mechanism. These prompt variants reuse exactly the same numerical observations and forecast queries.

Each task requests ten counterfactual forecasts spanning five intervention magnitudes and horizons one and three. Response error is computed from the eight nonzero-shock responses after subtracting the corresponding zero-shock forecast at each horizon. Evaluation crosses two domains, two response structures, three hint conditions, four target-evidence depths, and 30 episodes, for 1,440 prompts per checkpoint.

The primary contrast is population-prior minus episode-matched response MAE under the correct pathway description. Positive values favor episode-matched training. The four evaluation cells measure performance in distribution, under domain shift, under structure shift, and under both shifts jointly.

All reported endpoints use round 300 and five stochastic forecasts per prompt at temperature 0.7. Each forecast draw is scored separately before averaging within a prompt. Invalid forecasts receive an error equal to the observable range of the corresponding domain.

D.2 EIGHT-SEED TRANSFER RESULTS

Qwen3-4B and Qwen3-8B are each evaluated with eight training seeds per reward arm, using seeds 42–49. The data, 300-round endpoint, five-draw decoder, and transfer estimands are shared across the two reward conditions. The hierarchical bootstrap first resamples training seeds and then paired evaluation prompts within each selected seed. Brackets give 90% hierarchical bootstrap intervals. We additionally summarize seed-level uncertainty with a two-sided wild-cluster bootstrap with Webb weights and an exact one-sided sign test.

The eight-seed roster is a precision extension rather than the prospectively fixed seed count of the initial study. The endpoint, training data, decoder, estimands, and added-seed analysis procedures were fixed for the extension before the added endpoints were available. Because the decision to increase the seed count followed an earlier analysis, we interpret the eight-seed results as precision estimates rather than as a fully prospective confirmatory test.

Table 15: **Eight-seed population-prior minus episode-matched response MAE.** Positive values favor episode-matched training, and brackets give hierarchical 90% intervals. All cells use the correct pathway description and average the four evidence depths equally.

Evaluation cell	Qwen3-8B	Qwen3-4B
In distribution	.060 [.023, .103]	.067 [−.037, .171]
Domain shift	.110 [−.006, .242]	.087 [−.061, .232]
Structure shift	−.003 [−.046, .035]	.052 [−.055, .159]
Joint shift	.126 [.018, .242]	.136 [−.005, .275]

For Qwen3-8B, episode-matched training has lower point-estimate error in distribution, under domain shift, and under joint shift, while the structure-shift contrast is approximately zero. The in-distribution and joint-shift contrasts have hierarchical intervals excluding zero; the domain-shift and structure-shift intervals include zero. The in-distribution contrast is also supported by the wild-cluster test, $p = .0089$, and by eight positive seed-level effects out of eight. The corresponding one-sided exact sign-test value is $p = .0039$.

The Qwen3-8B joint-transfer contrast is .126 with hierarchical interval [.018, .242]. The interval excludes zero, but the wild-cluster test gives $p = .091$ and six of eight seed-level effects are positive, so the estimate remains compatible with a small advantage from episode-matched supervision rather than a large one. The domain-shift interval likewise includes zero, while the structure-shift estimate is centered near zero.

For Qwen3-4B, all four hierarchical intervals include zero. Its point estimates are positive in every evaluation cell, but the eight-seed data do not distinguish these effects from zero under the reported hierarchical intervals.

Taken together, the eight-seed results provide evidence that episode-specific supervision improves Qwen3-8B performance on the response relationship seen during training. They provide weaker evidence that this advantage transfers under a new domain or a new response mechanism alone; under joint shift the hierarchical interval excludes zero, but the wild-cluster test and the six-of-eight seed count do not support a clear effect. The results therefore distinguish learning the episode-specific training relationship from robust transfer of that relationship under structural change.

D.3 EXPLORATORY DIAGNOSTICS OF RELATIONSHIP LEARNING

The eight-seed experiment above evaluates whether episode-specific supervision improves transfer. The following diagnostics ask a narrower question: when transfer fails, is the underlying response relationship learnable, and does training make forecasts track episode-specific evidence rather than only a population-level target distribution? These analyses use separate experimental rosters and are not part of the primary eight-seed transfer comparison.

D.3.1 SUPERVISED LEARNABILITY AND ITS TRANSFER BOUNDARY

We first ask whether the response-structure task is learnable from direct supervision. This diagnostic uses a separate structure-selection dataset in which every prompt contains one direct and one mediated reference. Their horizon-one response magnitudes are matched, while their horizon-three persistence differs, so selecting the correct response form requires distinguishing temporal structure rather than immediate response magnitude. Targets are balanced equally between the two structures. Qwen3-8B is fine-tuned on 4,800 such Coin City prompts using simulator-correct forecasts as targets. The higher-rate supervised recipe uses learning rate 2×10^{-5} , one epoch, 300 updates, and four independent training seeds. Neither Coin Harbor nor arbitrary labels appears during training.

Across the four seeds, in-distribution Coin City forecasts identify the correct response structure on 97.9%–100.0% of prompts, with response MAE between .277 and .658. The same models remain near the binary-choice baseline when the relationship must transfer to arbitrary labels or Coin Harbor. City/arbitrary accuracy ranges from 39.6% to 50.0%, Harbor/semantic from 45.8% to 52.1%, and Harbor/arbitrary from 47.9% to 50.0%. Reference identification remains 100% throughout.

The supervised result establishes that the in-distribution response-form selection problem is learnable by this model and recipe. Its failure under arbitrary labels and domain transfer shows that successful in-domain structure selection does not imply inference of the response relationship from unfamiliar reference examples. This is not a controlled SFT-versus-RL comparison because the objectives, learning rates, and effective training budgets differ.

D.3.2 MECHANISM-FAMILY EVIDENCE USE UNDER COHERENT GAIN CHANGES

This diagnostic uses a separate Qwen3-4B mechanism-family training study with eight response mechanisms represented during training and four held out for evaluation. Episode-matched and shuffled-target models use the same mechanism-balanced prompts and training budget; the shuffled arm preserves the answer-vector distribution within compatible mechanism and evidence-depth groups while breaking the episode-specific pairing. We test whether trained forecasts track an episode-specific change in response strength. The diagnostic contains 16 paired episodes from each of 12 response mechanisms, comprising eight mechanisms represented during training and four held-out compositions or parameter-extrapolation mechanisms. Both mechanism-disclosed and mechanism-undisclosed conditions use the same episodes and nine target observations.

Within each pair, the target gain changes from the 20th to the 80th percentile of that mechanism’s gain distribution. Inputs, observation-noise draws, reference trajectories, baseline state, and mechanism remain fixed, while target observations and forecast outcomes are recomputed under the new gain. The disclosed description states the same gain range for both members and never reveals the realized gain.

Let Δr_i denote the simulator’s change in the eight-coordinate response vector between the high- and low-gain episodes, and let $\Delta \hat{r}_i$ denote the corresponding predicted change. We measure

$$E_{\Delta} = \frac{1}{8N} \sum_{i=1}^N \|\Delta \hat{r}_i - \Delta r_i\|_1, \quad \beta_{\Delta} = \frac{\sum_i \Delta \hat{r}_i^{\top} \Delta r_i}{\sum_i \|\Delta r_i\|_2^2}.$$

Exact tracking gives $\beta_{\Delta} = 1$, while a forecast invariant to the gain change gives zero.

The comparison uses five training seeds per arm of episode-matched and distribution-matched shuffled-target training.

Table 16: **Episode-specific gain tracking on seen and held-out mechanisms.** Entries are response-change MAE, with lower values better. Slope M and S are the tracking slopes for episode-matched and shuffled-target training. Positive shuffled-minus-matched contrasts favor episode matching. Brackets give 90% intervals from 2,000 seed-then-episode bootstrap repetitions.

Disclosure	Set	Matched	Shuffled	No change	Slope M	Slope S	Shuffled – matched
Disclosed	Seen	1.250	1.512	1.591	.323	.084	+.262 [+ .220, +.308]
Disclosed	Held out	.898	.874	.826	.122	.049	−.024 [−.069, +.019]
Undisclosed	Seen	1.115	1.488	1.591	.461	.107	+.373 [+ .326, +.418]
Undisclosed	Held out	.878	.862	.826	.238	.060	−.016 [−.055, +.020]

Episode-matched training improves tracking on mechanisms represented during training. The response-change MAE advantage is .262 with mechanism disclosure and .373 without disclosure, with both intervals excluding zero. The corresponding matched tracking slopes, .323 and .461, are also substantially larger than those of the shuffled control.

This advantage does not extend to the held-out mechanisms. The shuffled-minus-matched contrasts are −.024 and −.016, and both intervals include zero. Moreover, the episode-matched held-out errors exceed the .826 no-change baseline despite positive tracking slopes of .122 and .238. The models therefore respond to held-out gain changes without accurately tracking them. Improved accuracy on familiar mechanisms does not establish transferred episode-specific evidence use.

D.3.3 DISTRIBUTION-MATCHED SHUFFLED-TARGET CONTROL

The shuffled-target control addresses a limitation of the population-prior comparison in the primary experiment. Within compatible mechanism and evidence-depth groups, it preserves the complete

multiset of training answer vectors while reassigning those vectors across episodes. Prompts, training budget, and target distribution are therefore fixed while the episode-specific prompt–target correspondence is removed.

The prospectively fixed comparison uses five training seeds per arm. Under stochastic decoding with the mechanism undisclosed, the shuffled-minus-matched response-MAE contrast is $+.094$. Its 90% hierarchical interval is $[+.041, +.168]$, the wild-cluster test gives $p = .0138$, and all five seed-level contrasts are positive. The corresponding exact one-sided sign test gives $p = .0312$.

With the mechanism disclosed, the stochastic contrast is $+.155$, with hierarchical interval $[+.062, +.243]$. Four of five seeds are positive, while the wild-cluster test gives $p = .079$. The distribution-matched control therefore supports an episode-matching advantage in the frozen five-seed undisclosed condition, while the strength of evidence varies across disclosure and inference procedures.

A later sensitivity extension adds three training seeds to each arm. Because the final seed was added after the seven-seed result was known, these eight-seed estimates are descriptive rather than prospectively calibrated. On the undisclosed stochastic endpoint, the contrast decreases to $+.061$, with hierarchical interval $[+.001, +.125]$, wild-cluster $p = .101$, and six of eight positive seeds. The interval excludes zero only narrowly, the point estimate is smaller, the wild-cluster test does not reject, and the seed count is weaker than in the five-seed roster, so the extension does not strengthen the evidence beyond the prospectively fixed five-seed comparison.

Taken together, these diagnostics separate three notions of learning. A supervised model can learn semantic response-form selection in the training domain without transferring that selection to arbitrary labels or a new domain. Episode-matched training improves tracking of gain changes for mechanisms represented during training but not for held-out mechanisms. A distribution-matched shuffled-target control provides evidence that episode-specific prompt–target pairing can matter, while the strength of that evidence depends on the evaluation condition and seed roster. None of these results establishes general transfer of an induced response relationship to unseen mechanisms.

E EXPERIMENT 4: HISTORICAL-MARKET TRAINING AND HELD-OUT EVALUATION

Experiment 4 tests whether outcome-based training improves forecasts for events and question families absent from the fine-tuning data. Models receive a prediction-market question, its resolution rules, and the market-price history available by a fixed forecast cutoff, then predict the probability of YES. Training uses a Brier-score reward, and evaluation separates markets temporally and by event and question family.

E.1 SAMPLE CONSTRUCTION AND LEAKAGE CONTROLS

The source archive contains 1,788,703 market records. The selected datasets contain 1,736 training, 512 development, and 1,024 test markets, with YES rates $.311$, $.275$, and $.301$, respectively. Training contains 991 distinct events and 1,456 normalized question templates; development contains 292 events and 453 templates; test contains 493 events and 866 templates.

We include binary markets with literal Yes/No outcomes and an archived Boolean resolution inferred from an extreme final token price. A fixed domain filter retains political, governmental, macroeconomic, and geopolitical questions while excluding sports, entertainment, cryptocurrency, and other out-of-scope categories. The forecast cutoff is 30 days before the earliest available resolution, market-close, market-end, or event-end timestamp. Eligible markets require at least four distinct pre-cutoff trading dates, a latest available price no more than seven days old, and a latest YES price in $[.02, .98]$. Prompts contain at most the latest 14 pre-cutoff YES closing prices together with the question, resolution rules, and cutoff date. Outcomes, terminal prices, market status, terminal timestamps, volume, and selection metadata are withheld.

Training cutoffs precede July 1, 2025, development cutoffs fall between July 1 and December 31, 2025, and test cutoffs fall between January 1 and June 30, 2026. Development markets sharing an event, available series identifier, or normalized question template with training are removed, and the

same exclusion is then applied to test markets against both earlier splits. Numeric values and month names are normalized when constructing template keys. The final train–development, train–test, and development–test family intersections are all empty.

The archive provides timestamped price histories but not historical revisions to static question or resolution-rule text. We therefore treat those strings as time-invariant. The design prevents leakage through post-cutoff prices and explicit family reuse, but it cannot establish that historical events were absent from model pretraining.

E.2 TRAINING AND EVALUATION

Training uses the 1,736 unique markets in a deterministic 4,800-presentation schedule. Every market appears at least twice, and the resulting 300 rollout rounds and 2,400 optimizer updates match the training budget used elsewhere in the paper. Models are optimized directly for Brier reward,

$$r = 1 - (\hat{p} - y)^2.$$

Table 17: **Historical-market training and evaluation configuration.**

Quantity	Value
Models	Qwen3-4B-Instruct-2507, Qwen3-8B, Llama-3.1-8B-Instruct
Training markets	1,736 unique markets; 4,800 presentations
Training	300 rounds; seeds 42, 43, 44
Reward	$1 - (\hat{p} - y)^2$
Optimization	Dr. GRPO; AdamW; learning rate 10^{-6}
Rollouts	16 prompts per round; 8 responses per prompt; temperature 1.3
Development	512 markets
Held-out evaluation	1,024 markets; temperature zero; one forecast per market

Held-out evaluation uses Brier loss. The reported trained score averages the three seed-specific losses rather than averaging their forecast probabilities. Invalid forecasts receive Brier loss one. Pointwise 90% intervals use 5,000 bootstrap resamples of connected event/template components, preserving markets that belong to the same connected family. These intervals quantify held-out market-sample variation conditional on the three fitted models.

E.3 HELD-OUT RESULTS

All three model families improve relative to their corresponding untrained checkpoints on the 1,024 family-disjoint test markets.

Table 18: **Held-out historical-market forecasting.** Trained Brier scores average three training seeds. Negative trained-minus-base differences favor training, with 90% family-clustered bootstrap intervals.

Model	Base Brier	Trained Brier	Trained – base [90% CI]
Qwen3-4B	.1316	.1159	−.0157 [−.02355, −.00809]
Qwen3-8B	.1287	.1164	−.0123 [−.01825, −.00684]
Llama-3.1-8B	.1520	.1150	−.0370 [−.05095, −.02449]

Every Qwen3-8B training seed improves on its untrained base, and the same base-relative pattern holds for Qwen3-4B and Llama-3.1-8B. These results establish base-relative improvement on temporally later markets whose event and normalized question families were excluded from fine-tuning.

The latest pre-cutoff market price obtains Brier .11369. A logistic recalibration of its logit, fit using only the training split, obtains Brier .11232. The trained-minus-market interval includes zero for Qwen3-4B, +.0022 [−.00187, +.00645], whereas Qwen3-8B and Llama-3.1-8B have higher error than the market by +.0027 [+0.00038, +.00555] and +.0013 [+0.00006, +.00332]. Against the recalibrated market benchmark, Qwen3-4B’s interval also includes zero, +.0036 [−.00047, +.00784],

while Qwen3-8B and Llama-3.1-8B have higher error by .00409 [.00139, .00714] and .00269 [.00091, .00484], respectively. The historical experiment therefore establishes improvement relative to the models’ untrained bases rather than superiority to prediction-market forecasts.

E.4 SCALE COMPARISON

A separate 318-market holdout tests how the same training recipe behaves across Qwen3 scale. These markets have forecast cutoffs between January 1 and June 8, 2026, and their identifiers, events, available series, and normalized question templates are absent from the training, development, and original 1,024-market test sets. The holdout contains 227 event families and 288 normalized templates, and selection does not use outcomes or market probabilities.

Qwen3 checkpoints from 1.7B through 32B use the same training data, three seeds, 4,800-presentation schedule, reward, and development rule. Each base and trained checkpoint receives five non-thinking forecasts per market at temperature .7, top- $p = .8$, and top- $k = 20$. Forecast probabilities are averaged within each five-draw group before computing Brier loss. All Qwen3 draws parse successfully.

Table 19: **Qwen3 scale comparison on the 318-market holdout.** Trained means average seeds 42–44, and lower Brier is better.

Checkpoint	Base	Trained mean	Trained – base
Qwen3-1.7B	.2253	.1248	–.1005
Qwen3-4B-Instruct-2507	.1522	.1366	–.0156
Qwen3-8B	.1369	.1286	–.0083
Qwen3-14B	.1281	.1270	–.0012
Qwen3-32B	.1266	.1265	–.0001

The base-relative gain decreases as the untrained Qwen3 checkpoint improves. At 32B, trained-minus-base is $-.00007$ with a 90% interval of $[-.00039, +.00023]$. The 14B trained model has lower Brier than the 8B trained model by .00164 with interval $[-.00043, .00299]$, but the trained 14B and 32B checkpoints are already close in absolute error.

Contemporaneous Polymarket prices obtain Brier .12655 on this holdout. Trained 14B minus market is $+.00040$ $[-.00019, +.00122]$, and trained 32B minus market is $-.00003$ $[-.00008, +.00002]$. Both intervals include zero. Under this fixed training recipe, larger untrained checkpoints approach the absolute performance of the trained models, so increasing parameter count does not imply a larger training gain.

Llama-3.1-8B provides an architecture comparison on the same holdout. Its trained mean is .12643, compared with .16417 for the base, for a difference of $-.03774$ $[-.05482, -.02230]$. Its trained-minus-market difference is $-.00012$ $[-.00039, +.00009]$. The Llama result therefore reproduces the pattern of substantial base-relative improvement without establishing lower error than the contemporaneous market.

Hosted-model references. DeepSeek V4-Pro, Claude Opus 4.8, GPT-5.6, and Kimi K3 are evaluated on the same 318 markets as external performance references. Their Brier scores are .12660, .12927, .13335, and .12682, respectively, compared with .12655 for Polymarket. Provider sampling and reasoning configurations differ from one another and from the local checkpoints, so these values provide context rather than a controlled model-family or scale comparison.

E.5 PRICE ABLATION: FORECASTING WITHOUT MARKET HISTORY

The historical-market prompts include dated pre-cutoff prices, whose latest value is itself a strong forecast. To test how much trained performance depends on this information, we evaluate the same trained adapters after removing the entire price-history block while leaving the question, resolution rules, cutoff, and output format unchanged. Models are not retrained for this ablation.

We report the 1,024-market held-out evaluation for Qwen3-1.7B, Qwen3-14B, and Qwen3-32B. Each trained value averages seeds 42–44. The no-price evaluation uses one temperature-zero forecast per

market, matching the primary held-out evaluation. The training-set YES rate of .311 provides a fixed climatology benchmark with Brier loss .21042.

Table 20: **Removing market-price history eliminates the held-out accuracy of the trained models.** Brier loss is lower when better. Trained values average three seeds.

Model	No-price base	No-price trained	Trained with prices
Qwen3-1.7B	.2611	.2468	.1183
Qwen3-14B	.2381	.2413	.1147
Qwen3-32B	.2935	.2933	.1151

Without price history, every trained model performs worse than the .21042 climatology benchmark. Removing prices raises trained Brier from .1183 to .2468 at 1.7B, from .1147 to .2413 at 14B, and from .1151 to .2933 at 32B. Training also provides no consistent advantage over the corresponding no-price base models: the 1.7B difference is $-.01429$ with a 90% interval of $[-.02838, -.00191]$, the 14B difference is $+.00322$ $[+.00097, +.00554]$, and the 32B difference is $-.00020$ $[-.00301, +.00249]$.

The historical-market gains therefore depend strongly on having market-price history available at inference time. This ablation evaluates models trained with prices and then tested without them, so it does not establish whether training on price-free prompts could produce useful price-free forecasts.

F EXPERIMENTS 3–4: SHARED TRAINING CONFIGURATION

Experiments 3 and 4 use the same Dr. GRPO training implementation unless a study-specific section states otherwise. Table 21 reports the parameters needed to reproduce the shared optimization procedure.

Table 21: **Shared Dr. GRPO training configuration.**

Quantity	Configuration
Adaptation	Frozen base model with LoRA rank 32 and $\alpha = 64$ on attention and feed-forward projections; bfloat16; no quantization
Objective	Dr. GRPO with group-centered rewards and no group-standard-deviation normalization; PPO clip .2; importance-sampling cap 2.0; one PPO epoch; KL coefficient 0
Optimizer	AdamW; learning rate 10^{-6} ; $\beta_1 = .9$, $\beta_2 = .95$; weight decay 0; maximum gradient norm 1.0
Batching	16 prompts per rollout round; 8 sampled responses per prompt; 128 trajectories per round
Schedule	4,800 prompt presentations; 300 rollout-and-update rounds; 2,400 AdamW updates; constant learning rate with no warmup
Training decoding	Temperature 1.3; top- $p = 1$; no top- k restriction
Invalid outputs	Missing, malformed, or length-truncated required outputs receive reward zero
Training seeds	Experiment 3 primary Qwen comparison: seeds 42–49; Experiment 4 historical-market training: seeds 42–44

One reported training “round” denotes one rollout-and-update cycle over 16 prompts and eight sampled responses per prompt. With the implemented microbatching and gradient accumulation, each round performs eight AdamW updates, giving 2,400 optimizer updates over 300 rounds. Study-specific datasets, reward functions, evaluation decoding, and uncertainty procedures are described in their respective experiment sections.